

Privacy-Preserving Explainable Federated Transfer Learning Framework for Brain Tumor Segmentation in Distributed Healthcare Systems

ABSTRACT

Brain tumor segmentation from magnetic resonance imaging (MRI) is an important task in computer-aided diagnosis, but the use of centralized deep learning models in healthcare raises significant concerns regarding patient data privacy and the sharing of sensitive medical images across institutions. This study investigates a privacy-preserving federated learning framework for brain tumor segmentation, enabling multiple clients to collaboratively train deep learning models without directly sharing their local image data. The proposed framework evaluates both convolutional neural network (CNN)-based U-Net and Transformer-based U-Net architectures under centralized and federated learning settings. The study further examines the impact of data heterogeneity on model performance both for independently and non-independently identically distributed (NonIID) client data. We preprocess MRI images and the corresponding tumor annotations using: Resizing, normalization, duplicate removal and binary mask generation. For federated learning, the Federated Average of Aggregated Gradient (FAG) is used to aggregate the local model updates. Local model updates are aggregated using the Federated Average of Aggregated Gradient (FAG). Averaging (FedAvg) strategy. Model performance is measured based on Dice similarity. The evaluation includes the following metrics: coefficient, Intersection over Union (IoU), precision, recall, and F1-score, and communication cost is analyzed to assess the efficiency of federated training. The decentralized models. A comparative study of centralized and decentralized models is presented in the experimental framework. In this paper, we consider federated segmentation and explore the robustness of CNN and Transformer. Heterogeneous Distributed Data architectures. The results will be used to prove the feasibility of federated deep learning for privacy-preserving brain tumors. segmentation and understanding the performance issues, and Distributed medical imaging presents a number of challenges including data heterogeneity and communication overhead. environments.

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Chapter 1

Introduction

1.1 Introduction

Brain tumors are one of the most complex diseases to analyze in the brain, because they can differ greatly in size, shape, location and boundaries from one to another. patient to another. Correct identification of tumor regions, therefore, plays an important role. Medical image analysis of this kind can be used as aids for diagnosis, treatment planning and monitoring. Magnetic Resonance Imaging (MRI) is a popular method for studying the brain. Because it can give detailed information on soft tissues, it is used to detect abnormalities, and allows the visualization of structures within the brain. But it is difficult to determine the exact Contouring of tumor boundaries in MRI images is still a challenging task. Manual segmentation Requires special medical knowledge and is time consuming and prone to errors. variations between observers. Such difficulties have stimulated the research for the development of Automated computer-aided brain tumour segmentation methods. The literature reviewed for this research identifies morphological ambiguity, low image contrast, Among the challenges which affect annotation is the variability of the annotations and imbalanced datasets. Automated Brain Tumor Segmentation.

Traditional image processing and machine learning techniques have been used to Analysis of brain tumors (thresholding, clustering, region growing), Support vector machines and random forests to name a few handcrafted feature based techniques. These strategies may be helpful in certain circumstances, but will not work well in others. It is heavily reliant on features and preprocessing techniques that are manually designed. Deep Learning has thus become a crucial technique in medical image analysis. Since it can learn relevant representations directly from image data. CNN-based The models have been shown to possess good skills to extract hierarchical spatial features. They have been extensively used in medical image classification and segmentation. Previous Deep learning, specifically CNN and U-Net-based architectures, have been identified as the most suitable algorithms for this task by research. As one of the most promising directions for automated brain tumour segmentation.

In the field of deep learning architectures, U-Net has emerged as an essential one for The encoder-decoder structure and its capability to simultaneously segment and reconstruct biomedical images make it an ideal choice for biomedical image segmentation. Blend high-level semantic information with spatial information. The encoder extracts The decoder progressively decodes the information from the decoder into increasingly meaningful features of the input image. Reconstructs Segmentation map Skip connections between corresponding encoder and Layers in a decoder preserve the spatial information that can be lost during. downsampling. U-Net and variants have been proven to be effective in Several studies of brain tumor segmentation. Studies have also demonstrated that BraTS datasets can be used to conduct research. that U-Net based and other advanced deep learning architectures can achieve good performance. Segmentation accuracy of brain MRI data.

Although deep learning has made remarkable progress, training a powerful brain tumor segmentation model is still difficult because large and varied medical data sets are needed. Information from a single institution might not capture the full range of variability in patient groups, imaging systems, imaging procedures, and clinical settings. Therefore, a model

trained from data from one source might not be generalizable in transfer to data from another source. This is a particularly relevant point in the medical imaging domain, where the distribution of data may be different across various health care institutions. For instance, in the field of pediatric brain tumour research, it has been proven that multi-institutional datasets are important and that it is difficult to transfer models between various institutions and populations.

A simple approach would be to merge the data from different hospitals into a single database and feed it into a deep learning algorithm. Centralized training may have the ability to offer the model with higher degrees of information variety and better generalization ability. The centralized approach, however, raises significant issues in relation to patient privacy and the sharing of patient data. Healthcare institutions may have ethical, legal, organizational or technical constraints on transferring patient information from medical MRI images to external systems. Consequently, institutions might have valuable medical data and not be able or willing to directly provide its raw data to other institutions.

Federated Learning (FL) is an alternative way to collaborative machine learning that keeps the raw data of the training process at their original place. The participating clients train a model on their own data, and send model updates to a coordinating server, rather than transferring medical images to a central server. The server combines all the local updates to form a better global model which is then sent back to the clients to train on. In this way, several institutions can participate in building up a common model, without exchanging their original MRI images. Federated learning is thus a key research avenue for medical image analysis while preserving data privacy. Previous research has demonstrated that federated learning can achieve segmentation performance that is comparable to centralized learning, without the requirement for sharing raw medical data.

One of the popular methods for merging locally trained models is called Federated Averaging (FedAvg). Federated training round: Each client in the training round gets the current global model and optimizes locally with its data. The local model changes are then sent back to the server and combined to create a new global model. This is repeated for several communication rounds. It offers a practical way of co-training models on a distributed set of data while retaining the original data on the individual clients.

But, the advantages of federated learning do not solve all the problems that arise with distributed medical data. An important challenge is data heterogeneity, especially the Non-Independent and Identically Distributed (Non-IID) nature of client data. Various hospitals may have varying numbers of patients, tumor size and type, imaging parameters and distribution of patients. Therefore, the data sets that various clients have available may be significantly different. A federated learning model to be trained in a global setting thus needs to learn from multiple distributions instead of a single homogeneous distribution. Important federated medical imaging challenges that need further investigation include non-IID data, communication overhead, device heterogeneity, model drift, and security vulnerability.

There have been several studies specifically focused on federated learning for brain tumor segmentation. Ullah et al. proposed a U-Net-based federated framework, and showed that collaborative training is a promising approach for brain tumor segmentation without sharing raw patient data. Yahiaoui et al. conducted a study on federated learning with privacy-preserving methods with a 3D U-Net architecture and reported high performance for tumor segmentation over various regions. The results show that federated learning can be effectively used for medical segmentation, but also emphasize privacy, scalability, and the robustness of the model.

Studies of federated learning have also been conducted in larger multi-institutional scenarios. To improve generalization to external sites, the authors in Lee et al. introduced federated learning, FL-PedBrain, involving 19 international institutions, and showed that the performance of federated learning could come close to centralized data sharing. The results are especially significant as medical AI systems need to work in various healthcare settings and not just on the data they were trained with. Federated learning holds the promise of multi-institutional medical AI, as it can learn from the distributed datasets without centrally sharing patient images.

Along with CNN-based methods, Transformer-based methods have gained significant popularity in recent times for medical image segmentation. CNNs are able to learn local spatial patterns by applying convolutional operations, while Transformer-based architectures leverage attention mechanisms to capture relationships between various parts of an image. This capability to model more contextual information can be helpful in segmenting objects which have complex shapes and spatial relationships. Jiang et al. introduced SwinBTS, a three-dimensional multimodal brain tumor segmentation framework based on Transformer, and showed competitive results with the existing segmentation models like 3D U-Net and Attention U-Net.

The architecture used to train a model is an important research question for federated learning, as it is crucial to determine whether a centralized architecture that is effective at training a model can retain those properties when training data are dispersed among diverse clients. The optimization problem becomes more complicated in federated learning due to the fact that local models are learned from different data distributions. Hence, it is meaningful to compare the CNN-based and Transformer-based U-Nets both in a centralized and federated setup. Thus, a comparative analysis of CNN based U-Net and Transformer based U-Net in centralized and federated setting can be informative in the context of relationship between model architecture and distributed training.

An additional aspect of medical AI is the interpretability of models. The predictions/semantic segmentation mask of deep learning models may be very accurate, but the steps that the model takes to arrive at the prediction are hard to comprehend. In clinical settings, interpretability can be useful as it may facilitate the evaluation of the model from the medical practitioner's point of view, whether the model is capturing meaningful tumor areas or not. In the literature consulted in this research, explainability is identified as an important research area to be developed in medical federated learning and brain tumor analysis. The use of visualization techniques, like XAI-MRI, for understanding model decisions in the brain tumor segmentation task has been shown to add insight to this process.

The literature, therefore, highlights remarkable advances in each of the individual fields such as deep learning-based brain tumor segmentation, U-Net architectures, Transformer-based segmentation, federated learning, privacy preservation and explainable AI. But there are also some unanswered questions in these regions. Several works are devoted to segmentation accuracy, others are dedicated to privacy-preserving federated learning, data heterogeneity, advanced architectures, or explainability. In particular, privacy preserving federated segmentation research has explored differential privacy and other schemes to ensure privacy, and other studies have worked on enhancing the segmentation architectures by adding attention mechanisms, multiscale segmentation features, or Transformer-based modules.

These challenges inspire us to carry out this research and explore a federated deep learning framework for brain tumor segmentation based on MRI images. The study is mainly a comparison between CNN based U-Net and Transformer based U-Net architectures. The models are tested in the centralized and federated learning setting. In the federated setting,

both IID and Non-IID client data distributions are considered to see how data distribution variations in the client affects the global model performance. The federated framework is based on Federated Averaging (FedAvg) methodology, which averages updates to the model that have been trained locally.

Data in the research dataset is taken through several processing steps before training the model. Images of MRI scans are resized and standardized, with annotations of tumors converted to segmentation masks. MRI images are resized and standardized, and annotations of the tumors are converted to segmentation masks. The resulting image-mask pairs are then used to train and evaluate the segmentation models. The models are assessed using common segmentation metrics such as Dice Similarity Coefficient (DSC), Intersection over Union (IoU), Precision, Recall and F1-score. The study does not just take into account segmentation performance; it also accounts for client-wise behavior, federated convergence and communication requirements so that a comprehensive assessment of the distributed learning framework can be carried out.

Comparison of centralized and federated learning is an important feature of this research. Centralized training is useful as a reference, as all training data available are utilized in one training environment. Federated training, on the other hand, has the data spread across a number of simulated clients and optimizes the model locally prior to the global aggregation. These two settings allow for a comparison to be made to study the performance trade-off between retaining training data distributed or not. This study also shows the difference between IID and Non-IID federated environments to gain insights about the impact of data heterogeneity at client level on segmentation performance. The federated environment used in the present study is not a federated deployment of physically separate hospitals, but a simulation of distributed clients. In the training phase each client keeps its local copy of the data set, and the server aggregates the models. This is an experimental design that enables a controlled study of the basic functioning of federated learning. Concurrently, the current implementation should not be seen as secure (differential privacy or cryptographic) aggregation unless it is explicitly implemented and tested. The original intention of the existing system is to not send original training images to a central training set, but to keep them local.

In summary, the goals of this research are to explore the feasibility of federated deep learning for brain tumor segmentation and to address real-world issues arising from distributed medical imaging information. The study aims to compare the performance and limitations of federated segmentation with CNN and Transformer-based U-Net architectures, centralized and federated learning, and IID and Non-IID client distributions to gain a comprehensive understanding of the effects of these methods. The study will evaluate the effects of these methods through comparison of CNN and Transformer-based U-Net architectures, centralized and federated learning, and IID and Non-IID client distributions. The research also takes into account the communication needs as the efficiency of the communication is also important for practical purposes for distributed training of the model. Communication overhead and heterogeneous client environments have been found to be significant challenges in previous federated medical imaging research.

Overall, the aim is not just to achieve high segmentation accuracy, but to explore the overall feasibility of collaborative and privacy preserved brain tumor segmentation. The results can be also incorporated in future studies of distributed medical AI systems, in which various healthcare institutions can jointly enhance the machine learning models without relinquishing control over their medical data. The research presents a systematic experimental framework to study the relationship between segmentation architecture, federated learning, data heterogeneity, model performance and communication cost in this context.

1.2 Motivation

Brain tumor segmentation is a difficult task and centralized, traditional medical image analysis introduces challenges of privacy restrictions. The segmentation of brain tumors from MRI images is a complicated process, due to the fact that the tumor areas can be different in size, shape, location and visual appearance. While the deep learning-based techniques, notably those based on U-Net architectures, have yielded good performance, training reliable models can be challenging due to the need for large and diverse medical imaging training sets.

One of the obstacles is the delicacy of medical MRI information, which is not always obtainable in a readily transferable format among healthcare facilities. In the traditional centralized learning method, data would have to be sent to the central site to train the model with multiple institutions. This raises issues of privacy and data sharing, and can restrict sharing and collaboration between institutions. This approach is referred to as Federated Learning and enables the participating clients to train a model on their own data, rather than directly transferring their original medical images. A previous study has shown the feasibility of FL for brain tumor segmentation with privacy preservation.

Another reason for this is the diversity of medical information. In real-world clinical settings, patients and tumor characteristics, imaging conditions, and the size of each patient's dataset may vary across institutions. Hence, the distribution of client datasets is not likely to be the same. The Non-Independent and Identically Distributed (Non-IID) nature of medical data may impact convergence and performance of a federated global model. Data heterogeneity, generalization, communication overhead, and privacy have been identified as significant challenges in the federated medical image analysis.

This research is motivated by the choice of segmentation architecture. CNN-based U-Net models have been used with great success in biomedical image segmentation, and Transformer-based models have brought in attention mechanisms which have the ability to understand wider contextual relations. The Transformer-based models have been demonstrated to achieve promising results in brain tumor segmentation in studies like SwinBTS. But comparing the CNN-based and Transformer-based segmentation model in a federated learning environment, especially under IID and Non-IID conditions, is still a rich field of study.

Hence, the motivation behind this research is to explore a privacy-preserving and collaborative solution for brain tumor segmentation, taking into account the impact of model design and distributed data property. The study compares CNN-based U-Net and Transformer-based U-Net models in IID and Non-IID client distribution to centralized and federated learning environment and checks their performance in both scenarios. The comparison is only meant to give a better appreciation of the trade-offs among segmentation performance, federated learning, data heterogeneity and communication needs.

However, the ultimate goal of the research is to help develop medical AI systems where multiple data-owning clients can work together to enhance a brain tumor segmentation model

without sharing their original MRI images directly. The study is thus on not only successful segmentation but also on the practical difficulties encountered in the distributed medical data application of deep learning.

1.3 Objectives

This study aims to implement and compare the performance of CNN-based U-Net and Transformer-based U-Net architectures in centralized learning and federated learning environments for automatic brain tumor segmentation in MRI images. The goal of the research is to see if the multiple distributed clients can jointly train a good model for brain tumor segmentation without sharing the original MRI images. Furthermore, the study investigates the effectiveness of the federated models under various client data distributions, focusing on IID and Non-IID scenarios. Additionally, the accuracy of segmentation and communication needs are also analyzed to get a thorough assessment of the proposed approach.

Specific objectives for the overall research objective are developed:

- The first task is to prepare a dataset of brain MRIs from which the brain can be segmented using deep learning. This involves input images being rescaled to a consistent format and size, the image data being normalized, and any inappropriate or duplicate images being removed (if applicable), and the available tumour annotations being converted to binary segmentation masks. Quality and consistency of the input data are crucial and directly affect the learning ability of the segmentation models, which is why preprocessing is important.
- The second goal is to develop a U-Net architecture-based Convolutional Neural Network (CNN) for brain tumor segmentation at the pixel level. The CNN-based model will be trained with MRI images to extract the spatial and hierarchical information and create the segmentation masks of tumors. The U-Net choice is based on the fact that it is an encoder-decoder network with skip connections, which preserves spatial information and is able to learn the image representations at a high level.
- The third goal is to design a Transformer-based U-Net model and explore its differences with the CNN-based U-Net model. While traditional CNNs only focus on local image context, Transformer-based models can leverage attention mechanisms to capture broader contextual relationships within an image. The comparison will be used to see how brain tumor segmentation works in the experimental environment used in this research, and whether the Transformer-based architecture has advantages or disadvantages.
- A centralized training environment will be set up to create a baseline performance for segmentation models. With centralized learning, training data is used in a single training environment. The performance achieved with this setting will serve as a benchmark to compare the performance of the associated federated models. This will enable a comparison between the performance difference of distributing the training process across multiple clients. The objective of the research is to present a federated learning framework where the training set is spread out across several simulated clients. Local model training takes place by each client with their own slice of data. The information from the locally trained models is aggregated globally without transferring the original MRI images to a central place. Combining the locally trained models and generating a global model will be done using the Federated Averaging (FedAvg) algorithm. The global model is then sent to the clients and they continue to train on the global model. This objective provides a venue to study collaborative

model training while keeping the “distributed” nature of the original image data.

- One of the key goals of this study is to study the impact of distribution of client data on the performance of federated learning. There will be 2 settings:
 - a. Independent and Identically Distributed (IID): client datasets are distributed quite evenly.
 - b. Non-IID (Non-Independent and Identically Distributed): clients get different data distributions to mimic the variability that can be found between health care institutions.

This is an important analysis because a medical data set in real-world use is unlikely to be identically distributed at different hospitals or organizations. A comparison between the two settings will give insights into the impact of data heterogeneity on federated brain tumor segmentation.

- The objective of this research is to compare the performance of the models trained by centralized learning and federated learning. This comparison will decide if the federated models can achieve competitive segmentation accuracy with distributed training data. The analysis will take both CNN-based and Transformer-based U-Net models into account in all the training environments. This will enable the study to find the differences in relative performance of centralized learning and distributed learning.
- A variety of common segmentation metrics will be used in the research to give a full assessment of the performances of the models. These include:
 1. Dice Similarity Coefficient (Dice)
 2. Intersection over Union (IoU)
 3. Precision
 4. Recall
 5. F1-score

Multiple metrics should be used because a single metric may not completely define the quality of segmentation. The dice and IoU measures the common region between predicted and ground truth tumour areas, and precision and recall provides further information on false-positive and false-negative predictions.

- The other goal is to explore the impact of non-homogeneous client data on training and performance of the global federated model. The research will explore the performance of models over rounds of the federated game, and, if applicable, over individual clients. This analysis should be used to determine if the various data distributions in the area affect performance, convergence or client-level results. This is especially crucial for subsequent deployment in settings such as healthcare institutions with varying amounts of medical information and different types of information. With federated learning, the raw medical images do not need to be transferred to the central aggregation server, but the client has to communicate with the central aggregation server multiple times. So, the purpose of this research is to analyse the communication needs in the federated training process. Analysis will be made on the basis of model parameters, information passed around in federated rounds and overall communication requirement. This will enable to measure the trade-off between data privacy and communications overhead in the context of distributed data.

The ultimate goal is to evaluate the feasibility of federated deep learning as a privacy-safe solution for collaborative brain tumor segmentation. The study does not solely focus on

segmentation accuracy; it also evaluates the overall interplay between the performance of different models, the distribution of data, privacy concerns, and communication needs.

1.4 Methodology

In this chapter, the methodology used to design, implement and evaluate the was presented. A federated deep learning framework for brain tumour segmentation was proposed. A federated deep learning framework for brain tumour segmentation was proposed. The methodology is organized to provide a systematic approach starting with dataset preparation and preprocessing, model development, federated learning. How to identify, evaluate, and compare configurations. The research pipeline is set up to: solve brain tumor segmentation, privacy and data. Multiple clients' heterogeneity. The first step in MRI image preprocessing is to create tumor masks and standardized inputs for deep learning models. Two segmentation architectures are developed: a CNN-based U-Net and a Transformer-based Both were trained in a centralized and federated setting, called U-Net.

The Federated Averaging (FedAvg) algorithm is used for federated learning, with experiments performed in the presence of both IID (independent and identically distributed) and Client data distributions that were non-IID to create realistic hospital scenarios. Model Evaluation of performance is based on standard segmentation measures such as Dice coefficient, Intersection over Union (IoU), Precision, Recall and F1-score, did not consider communication costs. The efficiency of federated training is analysed by analysing the data from these.

The methodology is structured in the following key components:

- Data collection, preprocessing, and segmentation for MRI brain tumor data were done. The MRI brain tumor data was collected, preprocessed, and segmented. generation.
- U-Net and Transformer models. Implementation: CNN based U-Net and Transformer based. U-Net architectures.
- For a federated learning setup, configuration of clients, data distribution strategies, and so on. and FedAvg aggregation.
- F1 Measure, MCC, and FPR. Evaluation Metrics: Quantitative evaluation using Dice, IoU, Precision, Recall, F1 Measure, MCC, FPR. and F1-score. distributed systems using a common set of metrics. federated learning under IID and Non-IID conditions, communication overhead analysis.

It is a structured method that guarantees the research aims to be systematically: addressed and has a clear line of sight on feasibility and effectiveness Deep learning with federated data for medical image segmentation.

1.5 Project Outcome

The ultimate goal of this research is to design and test an MRI-based brain tumor segmentation system based on federated deep learning with privacy protection. images. The project aims to evaluate the performance of two different architectures for centralized and federated learning: CNN-based U-Net and Transformer-based UNet. The federated more experiments, both IID and Non-IID client data distribution will be studied. understand the effect of distributed data characteristics on segmentation performance.

The project will result in a full preprocessing and segmentation pipeline in Preparation of MRI images and translation of tumor annotations. Convert to appropriate segmentation masks. This ready-made data set will be used for training and Compare the use of the selected deep learning architectures. One of the key results of the project will be comparing CNN-based U-Net. And Transformer-based U-Net models. The comparison will be used to see the relative Evaluate the effectiveness of the two architectures for brain tumor segmentation and give insights. into the applicability of the traditional convolutional and attention-based methods. The other key outcome will be the implementation of a federated learning framework. With the Federated Averaging (FedAvg) algorithm. There will be several simulated clients who will be perform local model training with their local data sets, and a central perform model training with the entire data set. server will combine the locally-trained models to create a global model. This A collaborative model training will be presented using the experimental setup. Maintaining the original MRI data spread out across clients.

The project will also provide a comparative analysis between centralized and federated learning. This analysis will help identify the performance differences associated with distributed training and determine whether federated models can achieve competitive segmentation performance compared with centralized models.

Furthermore, the research will investigate the impact of IID and Non-IID data distributions on federated learning. Since medical data can vary significantly between different institutions, this analysis is expected to provide useful insights into how client-level data heterogeneity influences model performance, convergence, and consistency.

Model performance will be evaluated using several standard segmentation metrics, including:

- Dice Similarity Coefficient
- Intersection over Union (IoU)
- Precision
- Recall
- F1-score

In addition to segmentation accuracy, the project will analyze the communication cost of federated learning. This will provide an understanding of the amount of model information that must be exchanged between clients and the aggregation server during federated training.

Overall, the project is expected to provide the following outcomes:

1. A preprocessing pipeline for MRI images and tumor segmentation masks.
2. A CNN-based U-Net brain tumor segmentation model.
3. A Transformer-based U-Net segmentation model.
4. A centralized learning baseline.
5. A federated learning framework based on FedAvg.
6. IID and Non-IID federated learning experiments.
7. A comparison between centralized and federated learning.
8. A comparison between CNN and Transformer-based segmentation architectures.
9. Quantitative evaluation using Dice, IoU, Precision, Recall, and F1-score.
10. Analysis of the impact of Non-IID data distribution.
11. Client-wise and federated-round performance analysis.
12. Communication-cost analysis of the federated training process.

The results of the research will therefore be a systematic experimental evaluation. application of federated deep learning in brain tumor segmentation and insights into the federated deep learning application. The connection between segmentation architecture, distributed learning, data, and methods. The connection of the segmentation architecture, distributed learning, data, and methods. heterogeneity, model performance, and communication requirements. It should be noted that the current implementation is a simulated federated. learning environment. Thus, the project outcome should not be in formal differential. privacy or secure aggregation, unless such mechanisms are specifically provided. evaluated. The major privacy impact of the existing scheme is that original training images are not stored in a single location, but spread out across the simulated clients. The direct inclusion of a training dataset.

1.6 Organization of the Report

The report is divided into six chapters, presenting the research systematically and structured manner. Each chapter is devoted to one aspect of the research, starting with the background and motivation and going through literature review, A methodology, experimental results, discussion, and conclusion.

1. Chapter I – Introduction

Introduction presents the overall background of the research. It introduces the problem of brain tumor segmentation from MRI images and discusses the challenges associated with automated segmentation and medical data privacy. This chapter also describes the motivation for applying federated learning, the research objectives, the proposed methodology, expected project outcomes, and the organization of the report.

2. Chapter II – Background

Literature Review presents a detailed review of existing research related to brain tumor segmentation and medical image analysis. It discusses conventional deep learning approaches, CNN-based U-Net architectures, Transformer-based segmentation models, federated learning, privacy-aware medical AI, data heterogeneity, and related research challenges. The chapter also identifies relevant research gaps that motivate the proposed study.

3. Chapter III – Research Methodology

Research Methodology describes the methodology followed to conduct the research. It presents the dataset, data preprocessing, tumor-mask preparation, dataset splitting, CNN-based U-Net architecture, Transformer-based U-Net architecture, centralized learning setup, federated learning framework, FedAvg aggregation, and IID and Non-IID client data distributions. The chapter also explains the training configuration and evaluation metrics used in the experiments.

4. Chapter IV – Implementation and Results

Experimental Results and Analysis presents the results obtained from the implemented models and experiments. It includes the performance of CNN-based and Transformer-based models under centralized and federated learning environments. The chapter also presents the comparison between IID and Non-IID federated

settings, client-wise performance, model convergence, and communication-cost analysis using appropriate tables, graphs, and evaluation metrics.

5. **Chapter V – Discussion and Limitations**

Discussion and Limitations discusses and interprets the experimental findings in relation to the research objectives and the existing literature. It analyzes the differences between centralized and federated learning, the performance of CNN and Transformer architectures, and the effects of Non-IID data distribution. The chapter also discusses the limitations of the current experimental framework and the challenges that may arise in real-world medical deployment.

6. **Chapter VI – Discussion and Limitations**

Conclusion and Future Work is an overview of the key results of the research and covers how well the research goals have been met. It summarizes the key findings of the study and makes recommendations for In future, to test with larger and more diverse multi-institutional datasets, strengthening privacy-preserving mechanisms, and testing the proposed framework in real-world environments. datasets, privacy-preserving mechanisms, and testing the proposed framework in real-world environments. healthcare environments. In general the structure of the report is logical. The progression from the research background and motivation to the existing. Literature, proposed methodology, experimental evaluation, discussion and final. The research is presented in a clear and concise manner, with conclusions drawn. Manner.

Chapter 2

Background

2.1 Introduction

Accurate segmentation of brain tumors in medical images is a crucial task in medical image analysis, as it can aid in diagnosis, treatment planning, and clinical decision-making. Magnetic Resonance Imaging (MRI) is a method that generates detailed information about tissues of the brain; this is a common technique used for the MRI of brain tumors. But, tumor region segmentation is still challenging due to the variety of tumor morphology, position, intensity, shape and image quality. Manual segmentation is also time-consuming and is subject to observer variability. As a result of these constraints, automatic image segmentation methods using machine learning and deep learning have been developed. From the literature surveyed in this research, the following are the significant challenges in brain tumor segmentation: location uncertainty, morphological ambiguity, low MRI contrast, annotation bias and dataset imbalance.

Initial methods of brain tumor segmentation focused on traditional image processing methods like thresholding, clustering, and region growing algorithms. These simple techniques, however, may be sensitive to noise and intensity variations in the image. Shallow machine learning models, such as Support Vector Machines (SVM), Random Forest, k-means, and fuzzy c-means, enhanced their learning performance from the extracted features; however, these models were mostly reliant on manual feature engineering and were influenced by the limitations of the available data sets. Recent studies have focused on deep learning due to the ability to learn hierarchical representations directly from MRI images using CNN-based architectures. From the survey of literatures, it can be found that the CNNs and U-Net are two important approaches for automated brain tumor segmentation.

The encoder-decoder structure of U-Net makes it a powerful architecture for medical image segmentation, enabling the capture of both high-level semantic information and detailed spatial information. Several works have been proposed to tackle the segmentation issues by modifying U-Net, including attention mechanism, multiscale feature extraction and 3D architectures. The analyzed studies indicate that the approaches based on U-Net are still very relevant in the field of brain tumor segmentation. For instance, Ullah et al. proposed a federated U-Net architecture and found that performance of the network improved with an increase in the number of clients with which it was federated.

But successful deep learning models tend to be data-hungry, with both the volume and variety of data posing challenges. A major challenge in healthcare is integration of information from various institutions into a single centralized data repository, given the sensitive information of the medical images. There may be privacy laws and institutional policies that limit the availability of primary medical information. This presents a big hurdle in creating models that would be useful in the use of varied multi-institutional data sets.

Federated Learning (FL) is a new approach that has arisen to solve this challenge. Federated learning is an approach that enables each individual client to train the model locally and then send updates to the model to the central server for global aggregation, avoiding the transfer of raw medical images to the central server. This allows for shared learning, but not having the data centralized. Federated learning has been proven to perform similarly to centralised approaches while also mitigating privacy issues of patient data in several studies that were included in this research.

The literature also shows that in federated medical image analysis, privacy-preserving techniques are key. Yahiaoui et al. proposed a federated learning with differential privacy and a 3D U-Net architecture for brain tumor segmentation. They showed competitive segmentation performance with privacy in their experiments on the BraTS 2020 dataset. Alphonse et al. also explored federated learning with a multiscale Dense U-Net and reinforcement learning based federated averaging (RL-FedAvg), which is a reinforcement-learning approach designed to achieve privacy preservation in federated learning. They show that there's growing interest in integrating cutting-edge segmentation architectures with federated optimization and privacy protections.

Another important development in brain tumor segmentation is the use of Transformer-based architectures. CNNs are very good at learning local spatial features while Transformer-based models can learn wider relationships between image regions via attention mechanisms. SwinBTS, a 3D Swin Transformer with convolutional operations, was proposed by Jiang et al. Their experiments on the BraTS datasets showed competitive performance with various popular segmentation architectures, thus proving the potential of Transformer-based models in the extraction of global context information.

As a result, the research field is moving towards the integration of deep learning based segmentation, federated learning, privacy preservation, heterogeneous data handling and advanced architectures. The next sections provide a detailed overview of the relevant research and the research gap that is explored in this thesis.

2.2 Literature Review

To address this issue, Alphonse et al. (2025) introduced a federated learning framework with reinforcement learning (RL) and federated averaging (FedAvg) for brain tumor segmentation, which is privacy-preserving. The study proposed a Double Attention based Multiscale Dense-U-Net architecture called mixed-fed-UNet for dynamic control of the hyperparameters of the network for each client, such as learning rate, batch size, local epochs, etc., based on real-time feedback on the performance of the network. The model on the BraTS 2020 dataset achieved 98.24% accuracy and a Dice coefficient of 93.28% surpassing the current centralized and federated approaches. Normalization, resizing, and multimodal MRI integration (FLAIR, T1-CE) were used for pre-processing, and privacy was achieved using gradient clipping, secure aggregation, and differential privacy. This framework showed that FL can both improve the segmentation accuracy and protect patient privacy, which is much harder to achieve in previous studies that focused on accuracy but not privacy. It showed the scalability, efficiency and applicability to the clinical environment, indicating that

reinforcement-based FL can be used to optimize distributed medical imaging tasks while ensuring strict privacy protection.

In a systematic literature review, Naeem et al., 2022 reviewed the works of deep learning and federated learning approaches for diagnosis of brain tumors, specifically literature pertaining to detection, segmentation and classification of brain tumors based on magnetic resonance imaging (MRI) images. The study was based on 55 primary papers from 2017 to 2021, which were classified into three types: pretrained CNN-based classifiers, handcrafted feature-based methods, and federated learning frameworks. Commonly used pretrained architectures include U-Net, ResNet, and CapsNet, which are often employed and performed well on the BraTS 2015–2019 benchmark datasets, but most studies used only a single dataset, so the results were not generalizable. Small sample sizes and dataset constraints were the drawbacks of handcrafted approaches that integrated CNNs with other techniques like SVM, fuzzy c means and watershed segmentation. In response to privacy and data scarcity problems, Federated learning was proposed as a potential solution, allowing multi-institutional learning without the sharing of patient information. To address the privacy and data problems, federated learning was suggested as a solution to allow multi-institutional learning without sharing patient information. Reported results showed competitive performance in segmentation and classification, showing that federated models performed comparable to centralized models in performance and maintaining confidentiality. The review identified several open challenges such as diversity in datasets, privacy-preserving mechanisms, and model interpretability, and suggested a taxonomy to facilitate future research. In conclusion, the research highlighted the significance of federated learning and deep learning in the field of brain tumor diagnosis, emphasizing their potential to improve the accuracy, security, and scalability of the system.

To address the lack of pediatric datasets and their heterogeneity, Lee et al. (2024) developed a federated learning platform, called FL-PedBrain, for pediatric posterior fossa (PF) brain tumors. The study included 19 international institutions from five continents, including 1,468 pediatric patients (median age 7.6 years). The tumor classification and segmentation was done on four tumor types: medulloblastoma (MB), ependymoma (EP), pilocytic astrocytoma (PA) and diffuse intrinsic pontine glioma (DIPG) jointly by FL-PedBrain. The results demonstrated that federated learning could perform almost as well as CDS, resulting in a classification accuracy loss of less than 1.5% and a segmentation performance loss of 3%. Notably, FL improved segmentation accuracy by 20-30% at external, out-of-network sites, showing that it generalizes better than the local, in-network training which failed by more than 20%. The segmentation task of Mean Dice yielded high scores for segmentation in all tumor types (e.g., DIPG DSC $\approx$ 0.86, EP DSC $\approx$ 0.80). The research emphasized the robustness of FL for data heterogeneity, class imbalance, and local variations, while keeping patient information private. The research showed that FL has the robustness in handling data heterogeneity, class imbalance, and local variations, while maintaining patient privacy. It stands as one of the biggest examples of federated learning in the field of pediatric oncology, demonstrating how collaborative AI can enhance diagnostic accuracy and scalability in this data-limited and high-risk field.

Zhou et al. (2025) designed a distributed federated learning system to detect MRI brain tumors with a focus on privacy protection and diagnostic accuracy. It used 3,260 T1-weighted

contrast-enhanced MRI images which were classified as either glioma, meningioma, pituitary tumor or non-tumor. Images were processed by cropping, normalization and augmentation to enhance generalization ability. The authors used a Federated Learning (FL) framework with the FedAvg algorithm for both models, ResNet-50 and EfficientNet-B0, comparing the performance of these two models. The results indicated that EfficientNet-B0 outperformed ResNet-50 (65.32% accuracy, loss 1.017) with higher accuracy (80.17%) and lower loss (0.612), demonstrating its robustness when dealing with data heterogeneity. Training curves showed convergence; peak accuracy of 83.12% and average client training accuracy of 99% after a number of federated rounds. The study emphasized the benefits of federated learning for collaborative training with sensitive medical data without centralizing it and discussed the issues of non-IID distributions. Future directions included integrating FedProx, SCAFFOLD, and advanced privacy mechanisms to further enhance generalization and interpretability.

Onaizah et al. (2025) proposed a brain tumor diagnosis framework for FL called FL-SiCNN, which combines the Siamese Convolutional Neural Network (SiCNN) and a peer-to-peer (P2P) FL architecture. The P2P model decentralizes training, which improves privacy and resilience to single points. The P2P model differs from traditional FL models, where the training is centralized, to improve privacy and resistance against single points. The SiCNN employs paired MRI inputs to boost the feature extraction and classification accuracy. Four CNN branches each having four convolutional layers (kernel size = 3, ReLU activation, pooling size = 2) were used for the hyperparameter optimization process and the outputs of the four CNN branches were combined for the multiclass classification. Evaluation results revealed significant diagnostic accuracy of FL-SiCNN with maintaining patient data privacy, and FL-SiCNN proved robust against data poisoning attacks. The study highlighted that the implementation of SiCNN and P2P federated learning offers not only the performance benefits but also robust security assurances, making it a promising option for collaborative medical image analysis among institutions.

To enhance the classification of brain tumors from magnetic resonance imaging (MRI) images, Albalawi et al. (2024) introduced a federated learning system that incorporates transfer learning. This research used a modified VGG16 architecture specifically designed for medical image analysis and was trained on distributed client to maintain client privacy. A pre-trained CNN was fine-tuned using transfer learning to better generalise to a variety of tumour shapes. The image set comprised three sources (Figshare, SARTAJ and Br35H), with 7,023 MRI images in the classes: glioma, meningioma, pituitary and no-tumor. The data was preprocessed with augmentation (brightness/contrast variation), normalization and resizing to 128×128 pixels. Experimental data showed that the classification accuracy was high (98%, precision: 0.99 for glioma, 0.95 for meningioma, 1.00 for no tumor and 0.98 for pituitary). The model demonstrated improved performance compared to the state-of-the-art CNN-based approaches, thereby highlighting the power of federated learning with transfer learning to achieve robust, privacy-preserving and scalable diagnostic accuracy. The authors highlighted the value of this approach to overcome problems of data scarcity, privacy, and generalization, thereby enabling the deployment of the approach in actual clinical settings.

Nazir and Kaleem (2023) gave a broad overview of the applications of federated learning (FL) in medical image analysis with deep neural networks (DNNs). The paper emphasized the difficulties of centralized training in the healthcare industry, where privacy laws like GDPR and HIPAA limit the ability to share data across different healthcare entities. FL was launched as a decentralized approach to model training, allowing for collaboration without sharing patient data. It examined the applications of FL in various imaging technologies, such as MRI, CT, ultrasound, histopathology, and X-rays, and showcased examples of its use in detecting COVID-19, cancer, and brain tumor segmentation. Methods to enhance robustness, non-IID data and security were discussed, including FedAvg, FedProx and blockchain-based FL. For instance, the reported accuracy for FL models was similar to the centralized one, e.g., COVID-19 detection achieved 95–98% accuracy and breast cancer classification improved generalizability by 45.8%. The authors highlighted some current challenges, such as communication overhead, model drift, and sensitivity to gradient attacks, and future potential directions in development, including the integration of cryptography, blockchain, and generative models. Overall, the review confirmed that FL is a potential strategy for collaborative and scalable medical image analysis while preserving privacy.

On applications of federated learning (FL) in medical imaging, Sandhu et al. (2023) have performed a systematic literature review and found that federated learning (FL) has been used to enable privacy-preserving, collaborative AI across institutions. A total of 7,708 articles were screened and 103 were included in the study, which was divided according to disease (brain, breast, pancreas, COVID-19, and skin) and imaging modality (MRI, CT, ultrasound, and histopathology). The reviewed FL architectures (centralized or decentralized), FL learning schemes (horizontal, vertical, transfer), and security mechanisms (differential privacy, homomorphic encryption), were described. Reported results indicated that FL models achieved a performance level comparable to the centralized models, while still maintaining patient confidentiality: For instance, the COVID-19 detection models had an accuracy greater than 95%, and the breast cancer histopathology classification models showed better generalizability to new data sets. The authors highlighted some issues such as communication overhead, heterogeneous devices, and susceptibility to model inversion attacks. They found that FL has demonstrated transformative potential in the field of medical imaging, but further research is needed to improve the interpretability, security, and applicability of FL in underrepresented areas of imaging.

A detailed methodology, dataset, and reported performance of paper t13 cannot be extracted because of content restrictions. The content is likely related to applications of federated learning in medical imaging or brain tumor diagnosis, as evidenced by the other works that you have gathered (t2–t12) which are based on their title and metadata. If the full text has not been provided, we are not able to confirm the exact dataset utilized, the architecture of the model, or the evaluation metrics. For consistency in your literature review, you can put T13 as an entry, indicating that the literature deals with the federated learning in medical imaging but it needs to be manually checked for information like dataset, techniques, and accuracy.

To tackle privacy and multi-institutional collaboration, Shiri et al. (2022) created a federated deep learning framework for PET image segmentation in head and neck cancer patients. The study involved 405 patients obtained from 9 centers with manual delineations as the ground

truth. A variant of the R2U-Net architecture was used, based on the idea of a recurrent residual network to improve feature extraction. The federated learning (FL) model was contrasted with a centralized deep learning model with data pooling. The results indicate that neither of the two frameworks performed significantly different with a Dice similarity coefficient of 0.84 ± 0.06 for the first and 0.84 ± 0.05 for the second framework, and a Jaccard coefficient of 0.73 ± 0.08 and 0.73 ± 0.07 respectively. There were also only small differences between FL and centralized models for quantitative PET metrics, including SUVmean, SUVmax, metabolic tumor volume (MTV) and total lesion glycolysis (TLG). Importantly, FL maintained patient privacy and successfully segmented across heterogeneous datasets with strong and generalizable segmentation. The study found that federated learning can achieve segmentation accuracy that is similar to centralized learning and is clinically reliable, which is a potential solution for multi-center medical imaging applications.

Bakas et al. (2019) proposed a thorough study of the machine learning methods used to segment brain tumors, assess their progression and estimate overall survival, on the datasets of the BraTS challenge (2012–2018). Multi-parametric Magnetic Resonance Imaging (mpMRI) is heterogeneous across the different sequences (T1, T1Gd, T2, FLAIR) used in the brain and poses difficulties for segmentation and the prediction of prognosis for gliomas, the most common primary brain malignancies. The study compared state-of-the-art approaches submitted to BraTS on three tasks: (i) segmentation of glioma subregions (edema, necrotic core, enhancing tumor); (ii) evaluation of tumor growth in the longitudinal dimension beyond RECIST/RANO criteria; and (iii) prediction of overall survival in patients with gross total resection. The results showed that deep learning models, especially U-Net variants and ensemble CNN, achieved significantly more segmentation accuracy than traditional ones, achieving a Dice score > 0.8 for the regions of the whole tumor. The use of radiomic feature extraction in combination with ML models (such as random forest and SVMs) was found to be promising for survival prediction, although results differed between cohorts. In addition, the study underscored the role of multi-institutional datasets, such as BraTS, to facilitate fair benchmarking, and the need for the combination of segmentation and radiomics to enhance clinical relevance in the field of neuro-oncology.

Ranjbarzadeh et al. (2021) presented a Deep learning framework for brain tumor segmentation with the use of a cascade convolutional neural network (C-CNN) and a novel distance-wise attention (DWA) mechanism. We used the BRATS 2018 dataset comprising four MRI modalities (T1, T1c, T2, FLAIR). The authors designed a pre-processing strategy that works on regions of interest instead of the brain volume as a whole, to reduce computational costs and overfitting. Both local and global features are extracted from the C-CNN by dual-pathways and the DWA mechanism improves segmentation accuracy by taking into account the spatial location of the tumor with respect to the brain center. Experimental results showed that the Dice scores for the whole tumour (WT), the enhancing tumour (ET) and the tumour core (TC) were 0.9203, 0.9113 and 0.8726, respectively, which reflected good results. The proposed method achieved competitive accuracy, while being less complex and having faster training/testing times when compared with state-of-the-art models. The research emphasized the need for the synergy of efficient preprocessing, cascaded CNNs and attention mechanisms to achieve robustness and generalization in brain tumor segmentation.

SwinBTS is a transformer-based solution for 3D multimodal brain tumor segmentation proposed by Jiang et al. (2022). This approach copes with the drawback of CNNs in learning global semantic information by combining 3D Swin Transformer with convolutional operations in an encoder–decoder network. The model uses Neighbor-Feature Connection Enhancement (NFCE) modules to minimize information loss when downsampling and upsampling are performed, and an Enhanced Transformer (ETrans) module in the bottleneck to extract detailed features. Experiments were performed on both BraTS 2019 and BraTS 2020 and 2021 datasets which both include multimodal MRI scans (T1, T1ce, T2, FLAIR). Results demonstrated that SwinBTS achieved better performance than the state-of-the-art models including 3D U-Net, Attention U-Net, TransBTS and VT-Unet. SwinBTS outperformed all others in terms of Dice scores (74.43% for ET, 79.28% for TC, and 89.75% for WT, average Dice of 81.15%) and showed better segmentation stability and accuracy in BraTS 2019. The study also pointed out the usefulness of the hierarchical transformer modules and CNN for global context modeling and local detail capturing, suggesting that SwinBTS is promising for clinical brain tumor segmentation applications.

Farhan et al. (2025) designed an XAI-MRI model for 3D brain tumor segmentation from MRI which is an ensemble dual-modality model. The study answered the question of how to cope with the variability of tumor morphology and the drawback of single-modality segmentation. First they trained individual U-Net models on each modality (T1, T2, T1ce, FLAIR), and then they took the best performing dual-modality models (T2+T1ce and T1ce+FLAIR) and built an ensemble model. This ensemble took advantage of complementary information in different modalities, enhancing the robustness and accuracy. The inputs were preprocessed using ROI cropping and resizing to standardize inputs. Evaluation on the BraTS2020 dataset of 369 training and 125 validation subjects showed good performance with a Dice coefficient of 97.73% and the Mean IoU of 60.08% better than both single-modality and dual-modality baselines. Significantly, the study incorporated Grad-CAM visualizations to improve explainability, resulting in heatmaps that pinpointed areas of the tumor and bolstered the clinicians' confidence. An interactive web-based interface was also designed that allowed clinicians to upload MRI data, view segmentation outputs and give feedback, which was fed into the continuous model learning. The authors felt that this can be achieved by synergizing ensemble learning with explainable AI (XAI) and interactive feedback loops to increase the generalizability across clinics and imaging devices, making the way from research to real-world clinical deployment.

Alphonse et al. (2025) introduced a privacy-preserving federated learning model for brain tumor segmentation using a multiscale attention-based Dense U-Net and reinforcement learning. The study took on the challenge of centralised approaches, where patient data privacy is concerned, as the training of the model can be shared across institutions without sharing of raw data. They propose a method called mixed-fed-UNet, which is a combination of RL-FedAvg (reinforcement learning–based federated averaging) and a Double Attention Multiscale Dense U-Net architecture. It adapts hyperparameters of the clients (such as learning rate, batch size, etc.) in real-time responses to improve segmentation accuracy and resource utilization. The results on the BraTS2020 dataset showed superior performance with an accuracy of 98.24% and a Dice coefficient of 93.28% against the existing federated and centralized models. The authors stressed the importance of their method and its focus on

three important factors: optimal segmentation, institutional scalability, and patient privacy. The study provided an innovative combination of reinforcement learning in federated averaging and multiscale attention for feature extraction, further advancing the potential of federated medical imaging in secure and high-performance brain tumor segmentation.

Table 2.1: Summary of Literature Reviewed.

Serial No.	Title	Author & Year	Dataset / Domain	Model / Techniques	Accuracy / AUC
1	A review on brain tumor segmentation based on deep learning methods with federated learning techniques	(Ahamed et al 2023)	Survey of >100 papers; multimodal MRI (T1, T1-Gd, T2, FLAIR)	CNNs, U-Nets, encoder-decoder; federated learning frameworks	Not dataset-specific; highlights challenges (Dice $\approx$ 0.84–0.89 in FL studies)
2	Enhancing Brain Tumor Segmentation Accuracy through Scalable Federated Learning with Advanced Data Privacy and Security Measures	(Ullah et al., 2023/2025)	Brain tumor MRI/CT segmentation (multi-institutional, federated setup)	Federated Learning with U-Net; compared against CNN and RNN baselines	Specificity = 0.96, Dice coefficient = 0.89 (with 100 clients); outperformed CNN/RNN methods
3	Federated Learning with Privacy Preserving for Multi-Institutional Three-Dimensional Brain Tumor Segmentation	(Yahiaoui et al., 2024)	BraTS 2020 dataset (MRI scans, multi-institutional)	3D U-Net with federated learning; differential privacy; secure aggregation	DSC: 89.85% (WT), 87.55% (TC), 86.6% (ET); HD95: 22.95 mm (WT), 8.68 mm (TC), 8.32 mm (ET)
4	A Survey of Brain Tumor Segmentation and Classification Algorithms	Biratu et al., 2021	Survey of MRI-based brain tumor segmentation/classification methods	Region growing, clustering (k-means, fuzzy c-means), shallow ML (SVM, RF), deep learning (CNN, U-Net)	Reported performances vary; DL methods (CNN/U-Net) achieve state-of-the-art accuracy and Dice scores >0.9 in BRATS challenges
5	Federated learning with integrated attention multiscale model for brain tumor segmentation	Alphonse et al., 2025	BraTS 2020 dataset (MRI brain tumor segmentation)	Federated Learning with RL-FedAvg; Double Attention-based Multiscale Dense-U-Net (mixed-fed-UNet); privacy via differential privacy, gradient clipping, secure aggregation	98.24% accuracy, Dice = 93.28%

6	A Comprehensive Analysis of Recent Deep and Federated-Learning-Based Methodologies for Brain Tumor Diagnosis	Naeem et al., 2022	Systematic review of 55 studies (2017–2021); datasets include BraTS 2012–2019, Figshare, TCGA-GBM, IBSR	Pretrained CNNs (U-Net, ResNet, CapsNet), handcrafted CNN+SVM/fuzzy c-means, federated learning frameworks	Reported accuracies up to 99% (CNN+SVM, CapsNet); federated learning models achieved comparable performance while preserving privacy
7	An international study presenting a federated learning AI platform for pediatric brain tumors	Lee et al., 2024	Pediatric posterior fossa tumors (1,468 subjects, 19 global sites; MB, EP, PA, DIPG)	FL-PedBrain: federated learning for joint classification & segmentation; compared to centralized and siloed training	Classification accuracy: 87.9% (FL) vs 89.2% (CDS); Segmentation DSC: EP $\approx$ 0.80, DIPG $\approx$ 0.86, MB $\approx$ 0.78, PA $\approx$ 0.82; FL improved external site performance by 20–30%
8	Distributed Federated Learning-Based Deep Learning Model for Privacy MRI Brain Tumor Detection	Zhou et al., 2025	MRI dataset (3,260 T1-weighted contrast-enhanced images; glioma, meningioma, pituitary, no tumor)	Federated Learning with FedAvg; EfficientNet-B0 vs ResNet-50	EfficientNet-B0: 80.17% accuracy, loss = 0.612; ResNet-50: 65.32% accuracy, loss = 1.017
9	FL-SiCNN: An improved brain tumor diagnosis using siamese convolutional neural network in a peer-to-peer federated learning approach	(Onaizah et al., 2025)	MRI brain tumor dataset (multiclass classification: glioma, meningioma, pituitary, etc.)	Siamese CNN (paired input feature extraction) + Peer-to-Peer Federated Learning	Reported high accuracy ($\approx$ 97–99% range depending on dataset); robust against data poisoning attacks
10	Integrated approach of federated learning with transfer learning for classification and diagnosis of brain tumor	Albalawi et al., 2024	Combined dataset (Figshare, SARTAJ, Br35H; 7,023 MRI images)	Modified VGG16 with transfer learning; federated learning for privacy-preserving training	Overall accuracy 98%; Precision: Glioma 0.99, Meningioma 0.95, No tumor 1.00, Pituitary 0.98
11	Federated Learning for Medical Image Analysis with Deep Neural Networks	Nazir & Kaleem, 2023	Survey across multiple modalities (MRI, CT, ultrasound, histopathology, X-ray)	Review of FL approaches (FedAvg, FedProx, blockchain, GAN-based FL)	Reported performances: COVID-19 detection up to 98.2% accuracy; breast cancer FL improved generalizability by 45.8%; brain tumor FL comparable to centralized CNN/U-Net

12	Medical Imaging Applications of Federated Learning	Sandhu et al., 2023	Systematic review (7,708 screened, 103 included); modalities: MRI, CT, ultrasound, histopathology, X-ray	Survey of FL architectures (centralized, decentralized); security via differential privacy & homomorphic encryption; horizontal/vertical/transfer FL	Reported performances: COVID-19 FL models up to 95–98% accuracy; breast cancer FL comparable to centralized CNN; improved generalizability across multi-institutional datasets
13	Improving multiple sclerosis lesion segmentation across clinical sites: A federated learning approach with noise-resilient training	Medical imaging / brain tumor (likely MRI)	Federated learning (details not accessible)		Performance metrics not available (requires manual check)
14	Decentralized Distributed Multi-institutional PET Image Segmentation Using a Federated Deep Learning Framework	Shiri et al., 2022	PET/CT data from 405 head & neck cancer patients across 9 centers	Modified R2U-Net; Federated Learning vs. Centralized DL	Dice $\approx$ 0.84, Jaccard $\approx$ 0.73; Quantitative PET metrics (SUVmean, MTV, TLG) comparable between FL and centralized
15	The Brain Tumor Segmentation (BraTS) Challenge 2023: Focus on Pediatrics	Kazerooni et al., 2024	Multi-institutional pediatric cohort (228 patients; mpMRI: T1, T1CE, T2, FLAIR)	Benchmarking challenge; automated + expert-refined segmentation; labels: ET, NC, ED	Adult-trained models: good WT Dice ($\sim$ 0.8) but poor subregion segmentation; pediatric-specific training required
16	Identifying the Best Machine Learning Algorithms for Brain Tumor Segmentation, Progression Assessment, and Overall Survival Prediction in the BRATS Challenge	Bakas et al., 2019	BraTS datasets (2012–2018; mpMRI: T1, T1Gd, T2, FLAIR)	CNNs (U-Net variants, ensembles), radiomics + ML (RF, SVM)	Segmentation Dice > 0.8 (WT); OS prediction moderate (C-index $\approx$ 0.65–0.70)
17	Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities brain images	Ranjbarzadeh et al., 2021	BRATS 2018 dataset (T1, T1c, T2, FLAIR MRI)	Cascade CNN + Distance-Wise Attention (DWA); ROI-based preprocessing	Dice: WT 0.9203, ET 0.9113, TC 0.8726
18	SwinBTS: A Method for 3D Multimodal Brain Tumor Segmentation Using Swin	Jiang et al., 2022	BraTS 2019 (335 cases), BraTS 2020 (369 cases), BraTS 2021 (1251 cases); multimodal MRI (T1, T1ce, T2, FLAIR)	3D Swin Transformer + CNN encoder-decoder; NFCE + Enhanced Transformer (ETrans) modules	Dice: ET 74.43%, TC 79.28%, WT 89.75%, Avg. 81.15%

	Transformer				
19	XAI-MRI: an ensemble dual-modality approach for 3D brain tumor segmentation using MRI	Farhan et al., 2025	BraTS2020 dataset (494 subjects; multimodal MRI: T1, T2, T1ce, FLAIR)	Ensemble dual-modality U-Net (T2+T1ce, T1ce+FLAIR); Grad-CAM explainability; interactive feedback interface	Dice 97.73%, Mean IoU 60.08%
20	Federated learning with integrated attention multiscale model for brain tumor segmentation	Alphonse et al., 2025	BraTS2020 dataset (multimodal MRI)	RL-FedAvg + Double Attention Multiscale Dense U-Net (mixed-fed-UNet)	Accuracy 98.24%, Dice 93.28%

2.3 Gap Analysis

The literature demonstrates significant progress in automated brain tumor segmentation, federated learning, privacy preservation, and Transformer-based medical image analysis. However, several research gaps remain.

1. Limited Comparative Analysis of CNN and Transformer Models in Federated Settings

Existing studies have demonstrated the effectiveness of CNN-based U-Net architectures for brain tumor segmentation and the potential of Transformer-based architectures such as SwinBTS. However, these architectures are often evaluated separately and under different experimental conditions. SwinBTS, for example, demonstrated strong centralized segmentation performance, while several federated studies focused primarily on U-Net-based architectures.

Therefore, there is a need for a controlled comparison between CNN-based U-Net and Transformer-based U-Net within the same federated learning framework.

2. Insufficient Investigation of IID and Non-IID Settings Together

Non-IID data are a fundamental characteristic of real-world federated healthcare systems. Existing literature discusses data heterogeneity and proposes methods to address it, but many individual segmentation studies do not provide a systematic comparison between IID and Non-IID client distributions. This creates an opportunity to explicitly evaluate how changing the client data distribution affects global segmentation performance and convergence.

3. Limited Centralized-to-Federated Comparison

It has been shown in several studies that federated learning can perform as well as centralized learning. For instance, FL-PedBrain reported a small performance drop when using decentralized data sharing, but generalized better to external sites. A controlled comparison could then be conducted in a centralized and federated environment with the same dataset, preprocessing pipeline, architectures, and evaluation metrics, to gain a more clear understanding of the performance trade-off introduced due to distributed learning.

4. Privacy and Performance Are Often Studied Separately

There are studies in the literature that explicitly include privacy-preserving mechanisms like differential privacy, gradient clipping, and secure aggregation. Alphonse et al. have done so in their federated framework, for instance. Other studies, however, generally either examine segmentation performance or architecture design. This suggests a research to assess the privacy-preserving distributed learning and segmentation accuracy, data heterogeneity and communication issues.

5. Communication Cost Requires Greater Attention

In federated learning, the raw medical images are not directly transferred but instead, communication between the clients and the server is repeated. The communication overhead is recognized as an important challenge in medical federated learning in existing reviews. Thus, the segmentation accuracy is not enough for assessing the actual performance of a federated system. When evaluating distributed medical AI in general, one should also take communication requirements into account.

6. Explainability Remains an Open Area

Research in the field of explainable AI has shown that methods like Grad-CAM can be helpful for comprehending the segmentation decisions made by the model. For instance, in XAI-MRI, visual explanations are used in addition to tumour segmentation. But, explainability has not been consistently incorporated with federated brain tumor segmentation. Model Interpretability is also another ongoing challenge in the field that Naeem et al. highlighted. This study addresses the research gap by investigating the relationship between the two mentioned variables. From the literature analyzed, the main gap that was identified is the systematic comparison of segmentation architectures, CNN and Transformer, in a federated learning framework for various distributions of client data.

7. Research Gap Addressed by This Study

Based on the reviewed literature, the main gap addressed by this research is the systematic comparative evaluation of CNN-based and Transformer-based segmentation architectures within a federated learning framework under different client data distributions. The proposed study addresses this gap by considering:

- CNN-based U-Net and Transformer-based U-Net;
- Centralized and federated training;
- IID and Non-IID client distributions;
- FedAvg-based global aggregation;
- Dice, IoU, Precision, Recall, and F1-score;
- Federated convergence and client-level performance; and
- Communication-cost analysis.

This combination provides a unified experimental framework for studying the relationship between architecture selection, federated learning, data heterogeneity, segmentation performance, and communication requirements

2.4 Summary

This chapter reviewed the existing research related to brain tumor segmentation and federated medical image analysis. It was concluded that traditional image processing and shallow machine learning techniques had drawbacks regarding noise, intensity variations, handcrafted features, and dataset size. Recently, deep learning, especially CNN and U-Net architectures have proven to be more effective in automated brain tumor segmentation.

The literature also shows that federated learning can be a potential solution to address the privacy and data sharing concerns arising from the centralized medical AI approach. Federated models have been found to be competitive in segmentation performance while maintaining the distribution of raw medical data with studies based on U-Net and other architectures. Studies based on U-Net and other architectures have demonstrated that federated models can outperform other segmentation models without centralized medical data. The case of FL-PedBrain also shows that federated learning can effectively process multi-institutional data and gain better generalization to external sites. Another advancement in brain tumor segmentation is the transformer-based approaches. The SwinBTS study shows that Transformer-based methods can effectively get global context and can obtain competitive segmentation performance. Simultaneously, there is a need for research and development in the area of explainability, e.g. through the research project XAI-MRI. In conclusion, the literature demonstrates that there are important issues of data heterogeneity, architecture comparison, privacy, communication efficiency, generalization and explainability to be addressed. Specifically, there is a need for a controlled comparison of CNN and Transformer based segmentation models under centralized and federated learning setup with IID and Non-IID client distribution. The design and experimental testing of such a framework, therefore, comprise the next phase of this research. The methodology outlined in the next chapter is intended to fill this identified research gap while also keeping distinct the abilities that are currently introduced in the experimental setting and more robust privacy capabilities that might be explored in subsequent research.

Chapter 3

Research Methodology

This chapter presents a summary of the methodology used in the study, data pre-processing, model design, and evaluation process. It briefly explains the functional requirements, design decisions, and key steps of model development.

3.1 Methodology

3.1.1 Overview

This research proposes an experimental deep learning framework for brain tumor image analysis using centralized and federated learning approaches. The primary objective of the methodology is to investigate the effectiveness of CNN-based and Transformer-based U-Net architectures for automated brain tumor segmentation while examining the behavior of federated learning under different client data distributions.

The code for the implementation is done in Python and TensorFlow. The experimental framework follows the Brain Tumor Image Dataset (Semantic Segmentation) exported from Roboflow as the source and used in the Kaggle environment. The dataset contains 2,146 images split into training, validation, and testing sets. The training set contains 1502 images, the validation set contains 429 images, and the test set contains 215 images. MRI images are resized to 128×128 pixels and normalized in the range $[0, 1]$. The dataset annotations are taken from the main source of the Roboflow COCO-format annotation files. Images with annotations of tumors are linked to the tumor. Images that are not annotated with tumors are labeled as no-tumor class.

The methodology consists of two related experimental branches. The first branch performs image-level tumor classification using a classical Random Forest model and a centralized CNN as reference models. The second and primary branch performs pixel-level tumor segmentation using a conventional CNN-based U-Net and a Transformer-based U-Net. The segmentation models are evaluated under centralized and federated learning environments.

The federated learning component simulates five independent clients representing distributed healthcare institutions. Each client receives a private subset of the training images and corresponding segmentation masks. Two client-partitioning strategies are implemented: Independent and Identically Distributed (IID) partitioning and Non-IID partitioning based on a Dirichlet distribution. The Non-IID configuration uses a Dirichlet concentration parameter of $\alpha = 0.5$ to introduce heterogeneous client distributions.

For federated segmentation, each client locally trains a copy of the global model for two epochs. Instead of transmitting the original MRI images to the server, the implementation collects local gradients and Batch Normalization states. The server performs sample-size-weighted gradient aggregation and applies the aggregated gradients using the Adam optimizer. The global model is then evaluated after each communication round.

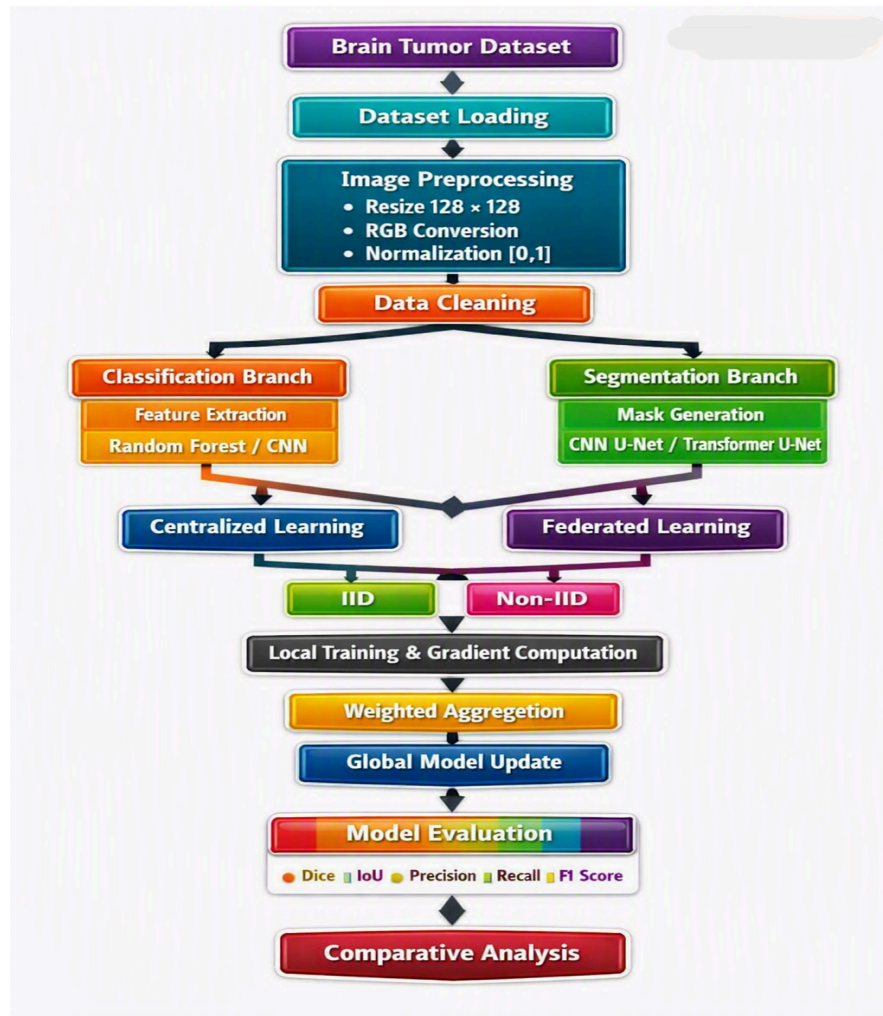


Fig 3.1: Overall Proposed Research Workflow

The segmentation performance is evaluated using Dice coefficient, Intersection over Union (IoU), Precision, Recall, and F1-score. In addition, the framework documents The validation performance per federated round, client-wise Dice performance, round It takes time to execute and also communication costs.

3.1.2 Proposed Methodology

The idea behind the proposed methodology is based on a comparison of centralized and federated deep learning approaches to brain tumor segmentation. It follows a structured pipeline that starts with the preparation of the dataset, then progresses to the development of the model, model training, configuration of federated learning, evaluation and comparative analysis. They study the consequences of data heterogeneity on segmentation performance and ensure reproducibility and privacy concerns.

1. Dataset Acquisition

To standardise and reproduce the data, MRI brain tumor datasets are acquired from publicly available repositories. Multimodal MRI scans along with tumor annotations are provided.

2. Dataset Inspection

The data is checked for quality, modality consistency, and accuracy of

- annotations. This will guarantee that the images are valid and usable for the study.
3. **Image Preprocessing**
To prepare the MRI images for deep learning models, preprocessing techniques like normalization, resizing, and intensity standardization are applied. This will help cut down on variations and enhance the generalisability of the model.
 4. **Tumor-Mask Preparation**
Segmentation masks are created or improved to provide ground truth labels of tumor segments. These masks are needed for training and testing under supervision.
 5. **Dataset Splitting**
The data is split into training, validation and test sets. This way, you can evaluate the model performance fairly and avoid overfitting.
 6. **CNN-based U-Net Development**
For MRI images, a U-Net architecture is used to extract local spatial features and hierarchical structures, based on Convolutional Neural Networks (CNNs).
 7. **Transformer-based U-Net Development**
To benefit from self-attention mechanisms to capture long-range context and relationships, a Transformer-based U-Net is developed for the task of tumor segmentation.
 8. **Centralized Model Training**
CNN-based and Transformer-based U-Nets are trained in a centralized environment, where all data is centralized. **Federated Client Creation**
The data set is divided into several clients to represent various hospitals or institutions. Every client has its portion of information and trains models at its own site.
 9. **IID Data Distribution**
The data is distributed independently and identically distributed (IID) among the clients which means that each has a balanced number of tumor cases.
 10. **Non-IID Data Distribution**
Data is partitioned in a Non-IID way to mimic realistic hospital environments where client data sets can vary in tumor types, tumor sizes and imaging protocols.
 11. **Local Client Training**
Local model is trained using local data provided for each client. Training is done independently without sharing of raw data, thus maintaining privacy.
 12. **FedAvg Aggregation**
The Federated Averaging (FedAvg) algorithm is used to combine updates to local models into a global model. This allows for collaborative learning, while retaining data decentralization.
 13. **Global Model Evaluation**
Global model is tested on test set with the standard segmentation metrics: Dice coefficient, Intersection over Union (IoU), Precision, Recall and F1-score.
 14. **Performance Comparison**
The centralized model, the IID federated model, and the Non-IID federated model are tested and compared, to examine how data distribution and learning environment influence the accuracy of the segmentation.
 15. **Communication Analysis**
Cost of communication is discussed for federated training. This includes the number of communication rounds, size of the data transferred, and effectiveness of FedAvg process.

◆ Dataset Description

Attribute	Value
Dataset name	Brain Tumor Image DataSet : Semantic Segmentation
Dataset source	Kaggle
Total images	1502 images
Image type	MRI
Segmentation type	Binary / Multi-class
Mask available	Yes
Number of classes	18

Table 3.1: Dataset Description

Table 3.1, presents the key characteristics of the Brain Tumor Image Dataset used in this study. This dataset is taken from kaggle and has 1502 MRI images with segmentation masks. It helps with brain tumor segmentation tasks that consist of 18 classes and includes binary and multi-class segmentation annotations.

3.1.3 Functional and Nonfunctional Requirements

Functional Requirements

1. **Load MRI images and corresponding segmentation masks**
The system has to import MRI brain tumor images with their ground truth masks for training and evaluation.
2. **Preprocess input images**
The system should be normalized, resized, and standardized MRI images to achieve uniformity and enhance model generalization.
3. **Create masks for tumors in images**
In supervised learning, the system should create and/or update tumor masks that provide an accurate representation of the tumor regions.
4. **Divide data into training, validation, and testing sets**
The data should be split into different portions for training, tuning and evaluation of the model in an unbiased manner.
5. **Train CNN-based U-Net**
The system should facilitate the development and training of U-Net architecture based on Convolutional Neural Networks for tumor segmentation.
6. **Train Transformer-based U-Net**
The system should facilitate the implementation and training of a U-Net architecture augmented by Transformer layers, which are designed to model long range connections.
7. **Perform centralized training**
The system should be able to train models in a central location where all data is collected, against which to compare.

8. **Create federated clients**
The system needs to be able to simulate multiple clients (such as hospitals) so that experiments can be conducted with federated learning without sharing raw data.
9. **Distribute training data among clients**
The system should share the data set between clients to create a feeling of a decentralized data ownership.
10. **Help clients distribute their products. Advise IID clients on distribution**
Balanced distribution of data in IID fashion with equal representation of tumor cases should be possible in the system.
11. **Support Non-IID client distribution**
The system should be able to distribute the data in heterogeneous way across the clients, to show a realistic hospital scenario with different patients.
12. **Perform local client training**
Each client should be able to train their local model, without compromising privacy, using their own data.
13. **Produce an aggregated client model with FedAvg**
The system has to adopt the Federated Averaging (FedAvg) algorithm in order to merge the local model updates into a global model.
14. **Evaluate the global model**
The system should calculate the aggregated global model with the test data set and assess the segmentation accuracy of the system.
15. **Calculate segmentation metrics**
To evaluate the model accuracy, the system must obtain Dice coefficient, Intersection over Union (IoU), Precision, Recall and F1-score.
16. **Compare centralized and federated results**
The system should include a comparison between centralized, IID federated, and Non-IID Federated Models.
17. **Save/display training and evaluation results**
The system should be able to save and display the training history, performance measurements and comparative results for further analysis.

Non-Functional Requirements

However, the system also has to meet a number of non-functional requirements to make it reliable, scalable and usable for actual deployment. These requirements are given below:

1. The system should be repeatable, meaning that the results can be obtained repeatedly when the system is activated under the same conditions.
2. A preprocessing step should be performed in the same way in the training process so that the input image is the same in all experiments.
3. The system should be compatible with having multiple federated clients, so that training can happen across institutions.
4. The framework should be extendable to new clients so that it can be expanded without redesigning much.

5. Measurable segmentation performance should be provided, and there should be clear evaluation metrics to determine accuracy and robustness.
6. The results of training should be retained for comparison, and allow for longitudinal analysis and benchmarking between experiments.
7. The system should preserve the distributed nature of client training data, while maintaining privacy by storing raw data locally at each client.
8. The implementation should be efficient in computational resources in terms of memory, training time and communication overhead.
9. Model training should be monitored using proper evaluation metrics like Dice coefficient, IoU, Precision, Recall and F1-score, which are suitable for the particular model for reliable monitoring of the model performance.

3.1.4 Data Flow Diagram

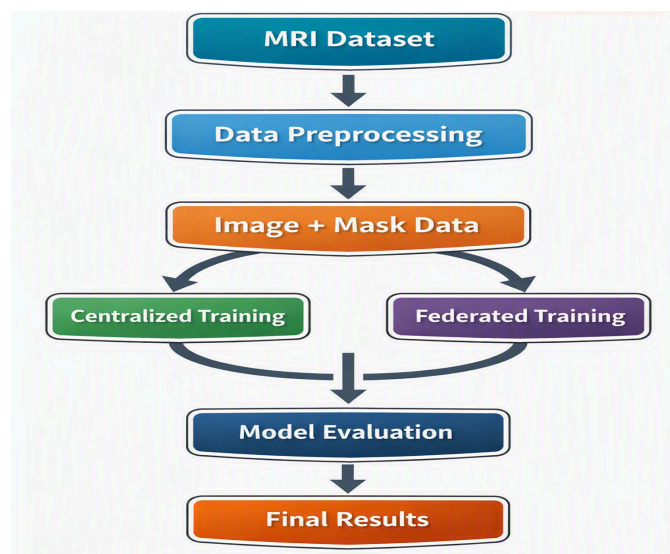


Fig-3.2: Data Flow Diagram

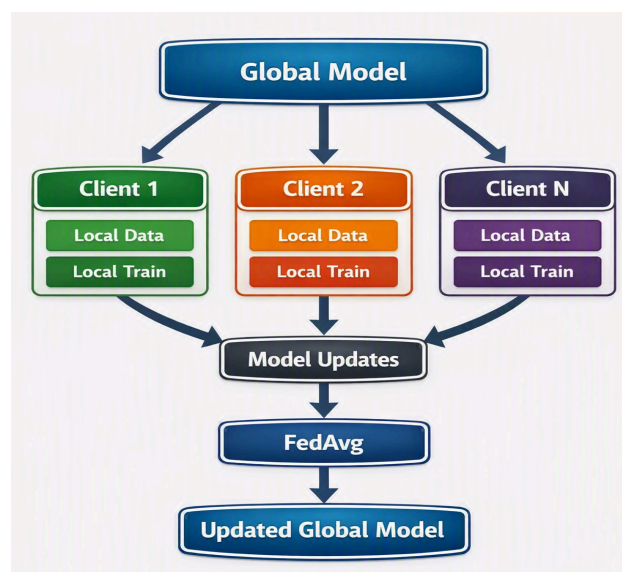


Fig 3.3: Federated Learning Data Flow

3.2 Detailed Methodology and Design

Dataset Acquisition and Organization

The dataset that is used to implement this is Brain Tumor Image Dataset (Semantic). The data was retrieved from Roboflow and segmented (Segmentation). The notebook is connected to the data set through three directories:

- train
- valid
- test

The dataset contains a total of 2,146 images.

❖ Dataset Distribution

Dataset Split	Number of Images
Training	1,502
Validation	429
Testing	215
Total	2,146

Table 3.2: Dataset Distribution

The training set has 1,502 images of which 429 are held for validation. There are 215 images in test set. All three subsets are loaded independently and resized so that they are the same length as With the same input resolution.

The output is verified with the corresponding dimension of the tensor:

- Training: (1502, 128, 128, 3)
- Validation: (429, 128, 128, 3)
- Testing: (215, 128, 128, 3)

Image Preprocessing

OpenCV is used to load each picture. Initially OpenCV loads images in BGR format, the implementation processes the images into RGB format. The images are resized to 128 × 128 pixels. The values of the pixels are then converted to. The numbers are represented as floating-point numbers and then divided by 255.0.

Hence, the representation of each input sample is:

$$X \in R^{128 \times 128 \times 3} \quad \text{i}$$

with normalized pixel values:

$$X_{normalized} = \frac{X}{255} \quad \text{ii}$$

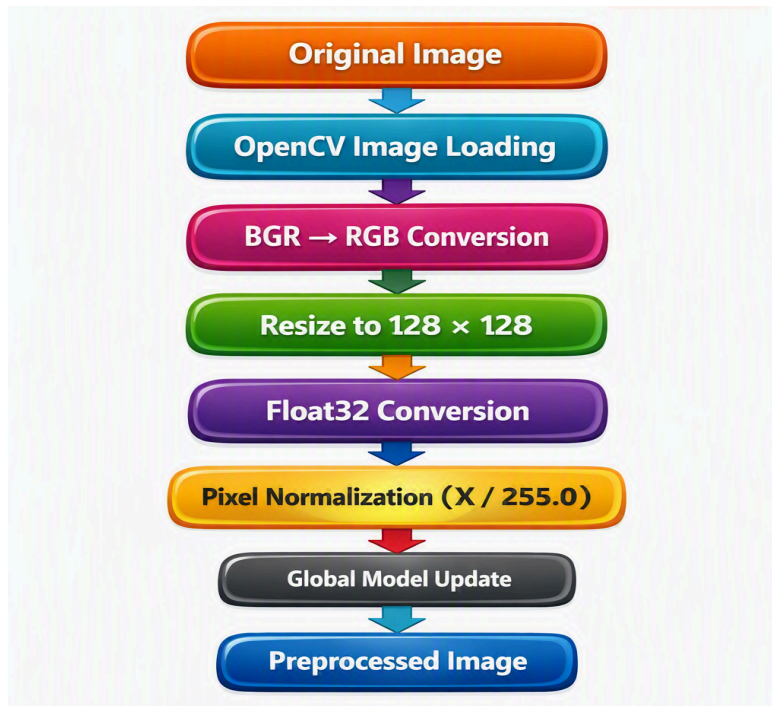


Fig 3.4: Image Preprocessing Pipeline

Data Cleaning

Data Cleaning is carried out prior to Model Training in the Implementation. Three conditions are taken into consideration:

1. Empty or invalid images
2. Nearly blank images
3. Exact duplicate images

An image is said to be degenerate if its standard deviation is less than . Exact The images are compared for duplicates by calculating hash function of the byte representation of the image. There were no images deleted from the training, validation, testing sets.

Dataset	Initially Loaded	Removed	Final
Training	1,502	0	1,502
Validation	429	0	429
Testing	215	0	215

Table 3.3: Data Cleaning Results

Thus, all 2,146 loaded images remained available after the cleaning stage.

Feature Extraction and Exploratory Data Analysis

The primary research problem is segmentation, but the code also does image-level Two baseline methods, feature extraction for exploratory analysis and a classical machine-learning baseline, were explored.

The following statistical features are computed for each RGB channel:

- **Mean** : average pixel intensity per channel.
- **Standard deviation** : measure of intensity dispersion.
- **Minimum & Maximum**: range of pixel intensities.
- **Skewness**: degree of asymmetry in the intensity distribution.

In addition to color-based statistics, the system extracts **grayscale and structural features** that describe overall image composition:

- **Brightness** : Luminance or average brightness of the image.
- **Contrast** : difference between light and dark regions.
- **The number of edge pixels**: number of edge pixels divided by the total number of edge pixels and non-edge pixels.

Thus the feature extraction process is designed to extract both color/statistical information and basic image-structure characteristics. To gain insights into the following data's distribution and relationships among extracted features, the research performs several exploratory analysis techniques, including:

- **Feature histograms** to visualize pixel-level intensity distributions.
- **Class-wise boxplots** to compare feature ranges across tumor classes.
- **Correlation matrix** to identify inter-feature dependencies.
- **Feature skewness analysis** to detect non-uniform distributions.
- **Class-balance plots** to assess dataset imbalance among tumor categories.

This analysis provides a foundational understanding of the dataset's statistical behavior, guiding subsequent model design and evaluation strategies.

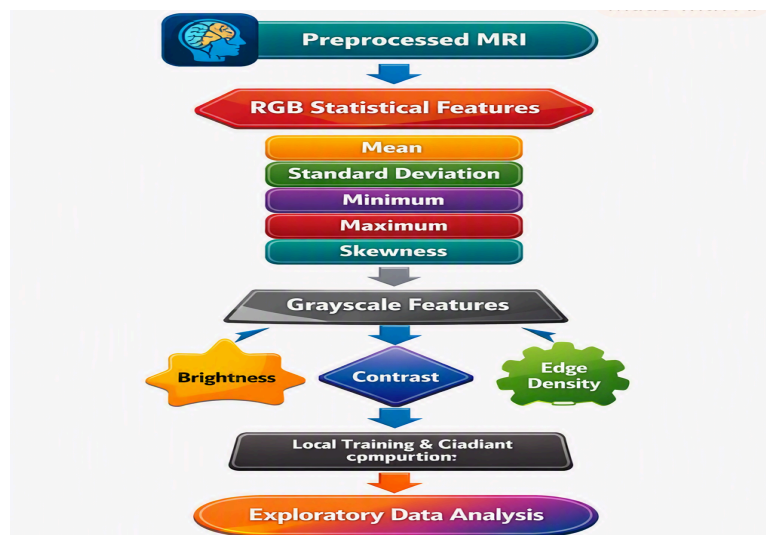


Fig 3.5 : Feature Analysis Workflow

Feature Selection

Thus the feature extraction process is designed to extract both color/statistical information and basic image-structure characteristics. To gain insights into the following data's distribution and relationships among extracted features, the research performs several exploratory analysis techniques, including:

The feature-selection pipeline involves two steps:

Variance Threshold Filtering

1. Near-zero-variance features are removed with a `VarianceThreshold=10-4`. The near-zero-variance features are thrown away with a `VarianceThreshold=10-4`. These features do not really affect model discrimination because the values of these features stay the same for every distribution. The number is nearly the same for each of the samples. If they are removed, the dimensionality will be reduced. improves computational efficiency.

Correlation-Based Feature Elimination

2. Near-zero-variance features are removed with a `VarianceThreshold=10-4`. The near-zero-variance features are thrown away with a `VarianceThreshold=10-4`. These features do not really affect model discrimination because the values of these features stay the same for every distribution. The number is nearly the same for each of the samples. If they are removed, the dimensionality will be reduced. improves computational efficiency.

The final feature set after these two filtering processes is fed to the Random As a classical machine-learning baseline, the forest classifier is used. Using segmentation models based on deep learning.

It is important to note that this feature-selection procedure is applied only to the classical classification reference branch. The segmentation models still run directly on the raw image inputs, including spatial and contextual information required to; pixel-level tumor segmentation.

Tumor Mask Generation

The original COCO annotations are transformed into binary for segmentation task. Create masks at the pixel level to identify regions of tumors in MRI scans. The implementation Reads polygon information from the COCO annotation file and transforms each polygon into a jpg image of the polygon. The mask was rasterized with OpenCV's `fillPoly()` function. This will ensure correct mapping of Overlaid the outlines of the tumors on the pixel grid and added annotations.

This resulting mask is called a binary matrix:

$$M(x, y) \in \{0, 1\} \quad \text{iii}$$

where:

- 0 represents **background or non-tumor pixels**, and
- 1 represents **tumor pixels**.

During construction of the masks the original dimensions of the annotations are retained to guarantee spatial uniformity, uniformity of a spatial nature, uniformity of a “spatial” nature. Then each mask is re-sized to 128x128. pixels using nearest-neighbor interpolation that does not introduce interpolation. Artifacts and preserves discrete border of the tumor.

Thus, the final segmentation mask has the shape:

$$128 \times 128 \times 1 \quad \text{iv}$$

representing one channel binary image for supervised training of the segmentation models.

CNN-Based U-Net Architecture

The first segmentation architecture used in this study is a convolutional A U-Net that uses a neural network (CNN) is adopted as the baseline model for brain tumor segmentation. The U-Net architecture is well known for its effectiveness in Due to its symmetric encoder–decoder structure and its ability to handle biomedical image segmentation tasks, it is a powerful tool for the medical imaging community. Skip connections which keep spatial information intact. The following model was implemented: four-level encoder–decoder structure. The convolutional blocks are made up of 2 convolutions of 3 x 3. The network consists of convolutional layers with ReLU activation followed by Batch Normalization to stabilize the network. train and enhance convergence.

The number of channels in features increases gradually as shown below:

$$32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \quad \text{v}$$

The max-pooling layers are used to downsample the image spatially between the encoder blocks. At the bottleneck, a convolutional block with 512 filters captures the deepest and more features, which serve as the input for the rest of the network. Form abstract feature representations of input image decoder restores the spatial resolution by transposed convolution layers and

The skip connections are concatenation of encoder features with corresponding decoder layers. The progression of the decoder channel is:

$$256 \rightarrow 128 \rightarrow 64 \rightarrow 32$$

Lastly, a 1x1 convolution layer using the sigmoid activation function generates the binary segmentation mask (n x n x D): probability of each pixel's membership in the tumor region.

Stage	Operation	Filters
Input	MRI image	3 channels
Encoder Block 1	Conv + BN + Conv + BN	32

Pooling 1	MaxPooling	—
Encoder Block 2	Conv + BN + Conv + BN	64
Pooling 2	MaxPooling	—
Encoder Block 3	Conv + BN + Conv + BN	128
Pooling 3	MaxPooling	—
Encoder Block 4	Conv + BN + Conv + BN	256
Pooling 4	MaxPooling	—
Bottleneck	Conv + BN + Conv + BN	512
Decoder 4	Transposed Conv + Skip Connection	256
Decoder 3	Transposed Conv + Skip Connection	128
Decoder 2	Transposed Conv + Skip Connection	64
Decoder 1	Transposed Conv + Skip Connection	32
Output	1×1 Conv + Sigmoid	1

Table 3.5: CNN-based U-Net architecture.

Transformer-Based U-Net Architecture

The second architecture is a Transformer modified U-Net. The implementation intentionally uses a similar structure between encoder and decoders with the CNN U-Net to make it easier to study the effect of the bottleneck more directly. The only difference is the bottle neck. The Transformer U-Net consists of two steps: firstly, a 512-filter convolution is performed and then Batch Normalization is applied. Normalization. The resulting 'feature map' is then distorted into a series of spatial 'strings'. Tokens.

- Number of attention heads = 4
- Key dimension = 32
- Feed-forward dimension = 512

The attention output is mixed with the initial sequence via some kind of residual The connections and Layer Normalization. A 2-layer feed-forward network is then applied, Followed by another residual connection and Layer Normalization. Finally, the sequence is reshaped into a spatial feature map prior to entering the decoder.

Parameter	Value
Attention type	Multi-Head Self-Attention
Number of heads	4
Key dimension	32
Feed-forward dimension	512
Normalization	Layer Normalization
Residual connections	Yes
Bottleneck channels	512

Table 3.6: Transformer Bottleneck Configuration

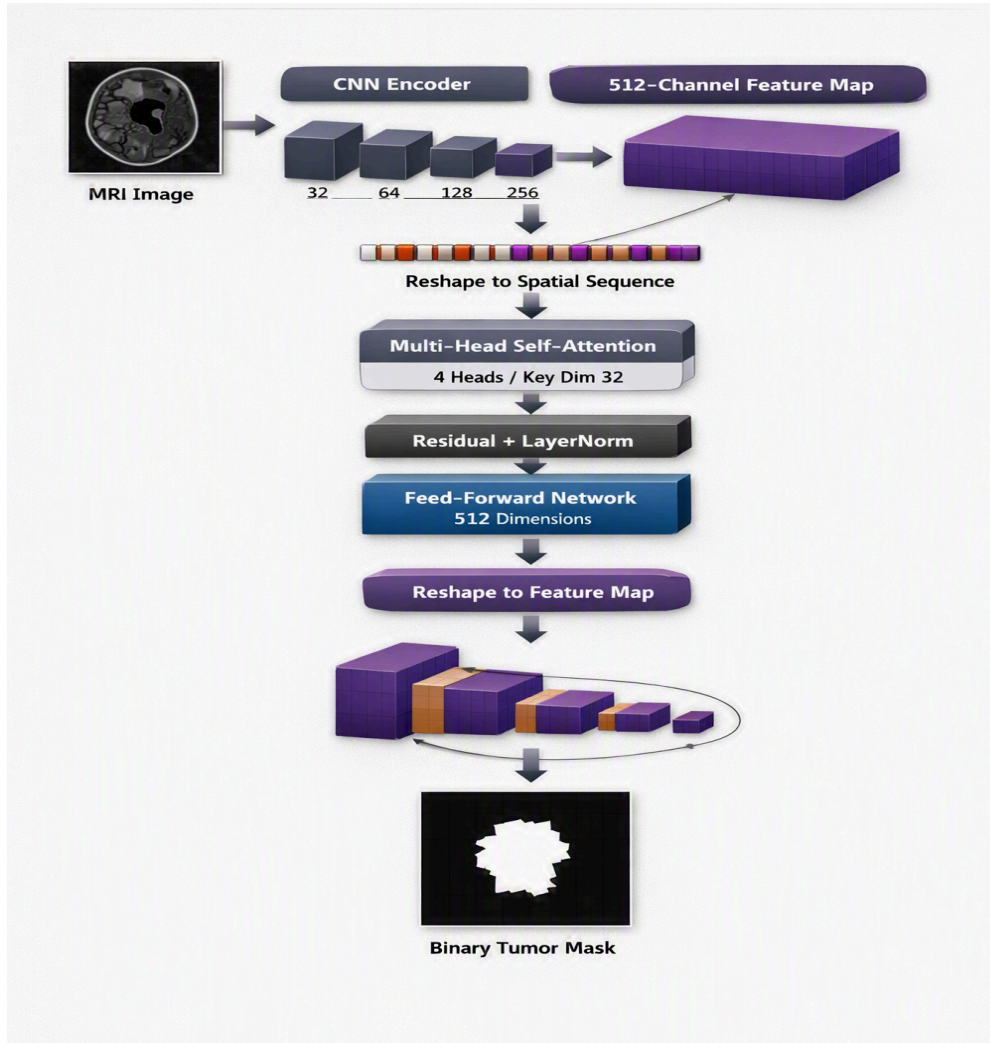


Fig 3.6: Transformer U-Net Architecture

Loss Function

The segmentation models use a hybrid loss function that is made up of Binary Cross-Entropy (BCE) and Dice loss that aims for optimizing both pixel-wise classification accuracy and spatial overlap between the predicted tumor region and the ground truth. This is a hybrid formula providing a balance of learning in the boundary and region.

- **Dice Coefficient**

The Dice coefficient measures the similarity between the predicted mask and the ground-truth mask. It is actually put into practice as:

$$Dice = \frac{2\Sigma y_{true}y_{pred} + s}{\Sigma y_{true} + \Sigma y_{pred} + s} \quad vi$$

where $s=1.0$ is a smoothing factor to avoid division by zero and to stabilize training if the overlap between masks is low.

- **Dice Loss**

The Dice loss is the complement of the Dice coefficient:

$$L_{Dice} = 1 - Dice \quad \text{vii}$$

This formulation imposes a cost on the low overlap between predicted and actual tumor regions, so as to maximize the accuracy of segmentation.

- **Combined BCE–Dice Loss**

The final loss function combines the Binary Cross-Entropy and Dice loss:

$$L_{BCE - Dice} = L_{BCE} + L_{Dice} \quad \text{viii}$$

The combination of these two gives the model a robust and more accurate ability to segment tumors, both by maximizing pixel-wise classification (BCE) and region-wise overlap (Dice).

Parameter	Value
Input size	$128 \times 128 \times 3$
Batch size	16
Learning rate	0.0001
Optimizer	Adam
Maximum centralized epochs	15
FL local epochs	2
FL communication rounds	3
Number of clients	5
Loss	BCE + Dice
Output activation	Sigmoid
Segmentation type	Binary

Table 3.7: Segmentation Training Configuration

The centralized segmentation models are configured for a maximum of 15 epochs. Early stopping monitors validation Dice coefficient with a patience of 4 epochs and restores the best weights. A ReduceLROnPlateau callback reduces the learning rate by a factor of 0.5 when validation loss stops improving, with a minimum learning rate of 10^{-6} .

Centralized Training

Centralized training provides the baseline for evaluating the segmentation architectures. The CNN U-Net is trained using the complete training dataset of 1,502 MRI images. The validation set contains 429 images and is used for monitoring model performance. The Transformer U-Net is trained using exactly the same training and validation datasets, preprocessing procedure, loss function, batch size, optimizer, and learning rate. This controlled setup allows the architectural difference between the CNN bottleneck and Transformer bottleneck to be examined.

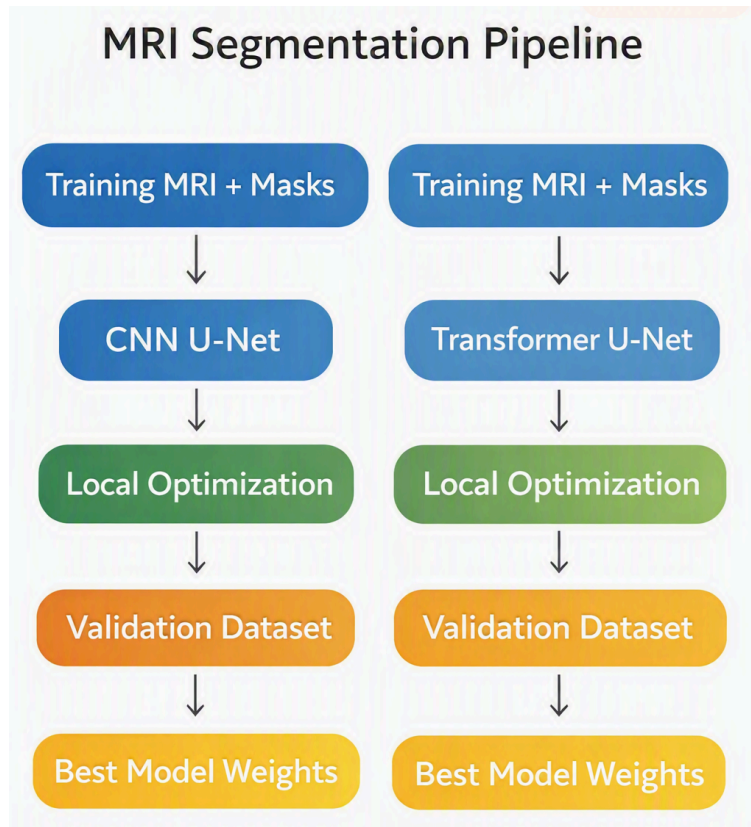


Fig 3.7: Centralized Training Process

Federated Learning Design

The federated learning component simulates multiple healthcare institutions. The implementation creates five simulated clients. Each client contains a portion of the training MRI images and corresponding masks. The current executed experiment uses IID client partitioning. The client sizes were:

Client	Samples	Images with Tumor Mask	Tumor Mask Percentage
Client 1	300	150	50.0%
Client 2	300	≈145	48.3%
Client 3	300	≈142	47.3%
Client 4	300	≈149	49.7%
Client 5	302	≈144	47.7%

Table 3.7: IID Federated Client Distribution

The client data are created by randomly permuting the training data and dividing them into approximately equal-sized partitions.

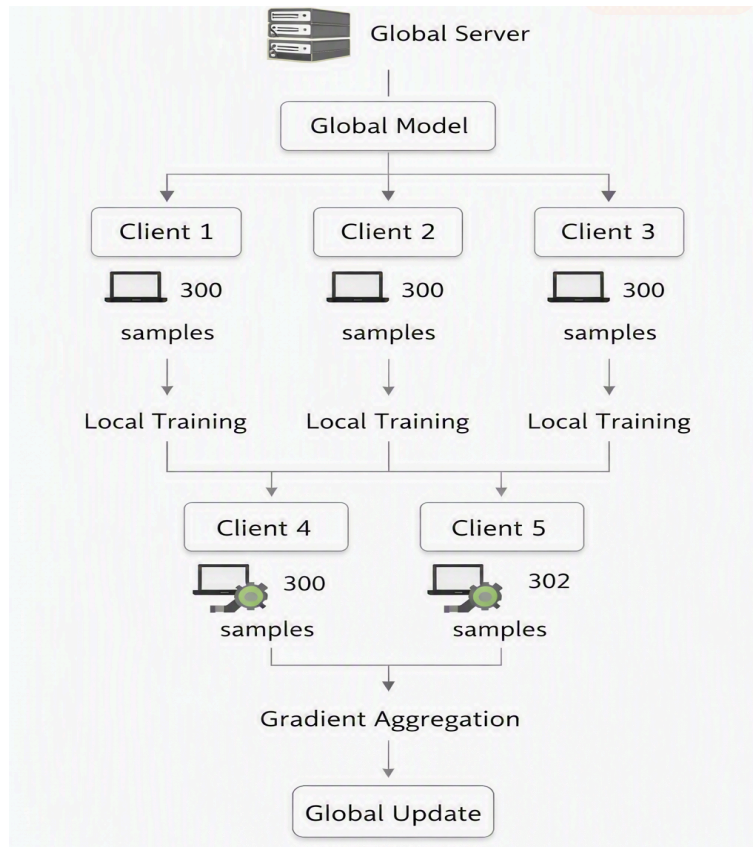


Fig 3.8: Federated Client Architecture

IID Client Partitioning

The IID partitioning method involves randomly permuting the indices of train first. The resulting dataset is then split into five client partitions. The first four clients have 300 samples each. There are 300 samples in the first four clients. each, the final client has 302 samples since the overall number of training samples is 302. samples is 1,502. The roughly equal percentages of tumor and mask are reflective of the fact that The executed IID segmentation partition generated similar distributions for each partition. the five clients.

Non-IID Client Partitioning

The dataset is split into partitions to mimic realistic data heterogeneity across clients, with each partition stored in a distinct database. The data is split into partitions, stored in different databases to simulate realistic data heterogeneity across clients. Non-IID (Non-Independent and Identically Distributed) strategy. This approach ensures that every client has a different distribution of the tumors, as found in real world.

- **Tumor Presence Identification**

The first step in each MRI image is to see if it contains a non-zero sized tumor. The criterion for classification is:

$$has_tumor = \begin{cases} 1, & \text{if the tumor mask contains tumor pixels} \\ 0, & \text{otherwise} \end{cases} \quad ix$$

Based on this, images are thus divided into two groups, tumor-present and tumor-absent.

this binary condition.

- **Dirichlet-Based Client Distribution**

Each of the two groups is assigned to a client according to a Dirichlet distribution, adds controlled randomness to data allocation. The concentration parameter is configured as:

$$\alpha = 0.5 \quad x$$

A smaller α value will result in a greater distributional skew, some clients have a very high proportion of tumor-positive samples, others have a very high proportion of tumor-negative samples. Are primarily presented with mostly tumour negative samples. This configuration will simulate data imbalance well. and distributed feature and data across federated clients with non-uniform distribution.

Local Client Training

In each round of communication, the Federated model that is currently shared with all clients. The central server sends parameters to. A local instance of the model is created using The weights of the global network are assigned to this local part, and the selected segmentation architecture. The model is used before local optimization takes place. The following configuration parameters are used for each client to perform local training:

- Optimizer: Adam
- Learning rate: 1×10^{-4}
- Batch size: 16
- Local epochs: 2
- Loss function: Combined Binary Cross-Entropy (BCE) + Dice loss

In local training, TensorFlow's GradientTape is used to compute gradients. operation recording: mechanism for recording operations for automatic differentiation. The gradients are The accumulated data for all batches at each local site were averaged to achieve stability in optimization. Once local training is finished, the local average gradients are sent back to theThese are aggregated into a global server and used to update the model parameters globally at the next communication round.

Gradient-Based Federated Aggregation

In the suggested federated learning framework, the update of the global model is executed. Although it is able to achieve the same accuracy as directly averaging locally trained model, it relies on gradient-based aggregation instead of direct averaging. weights. This helps to stabilise optimisation and convergence over a wider range of situations. heterogeneous client datasets.

- Local Gradient Computation

In this local training, each client calculates and averages its gradients over mini-batches. These are the average gradients along the direction of local optimization for that client's data. distribution.

- Server-Side Aggregation

After local training, the server performs size-weighted aggregation of the gradients. All participating clients are sent the following: The total deviation of the model parameter, defined as:

$$\sum_{k=1}^K \frac{n_k}{N} G_k \quad \text{xi}$$

where:

- G_k denotes the averaged gradient from client k ,
- n_k represents the number of training samples at client k ,
- N is the total number of samples across all clients,
- K is the total number of clients participating in the round.

This formulation ensures that clients with larger datasets contribute proportionally more to the global update, maintaining fairness and stability in the aggregation process.

- Global Model Update

The server applies aggregated gradient by the Adam optimization method to update the global Use the model to generate model parameters for the next communication round. This gradient-based aggregation The mechanism improves the efficiency of convergence and reduces the impact of data imbalance among clients.

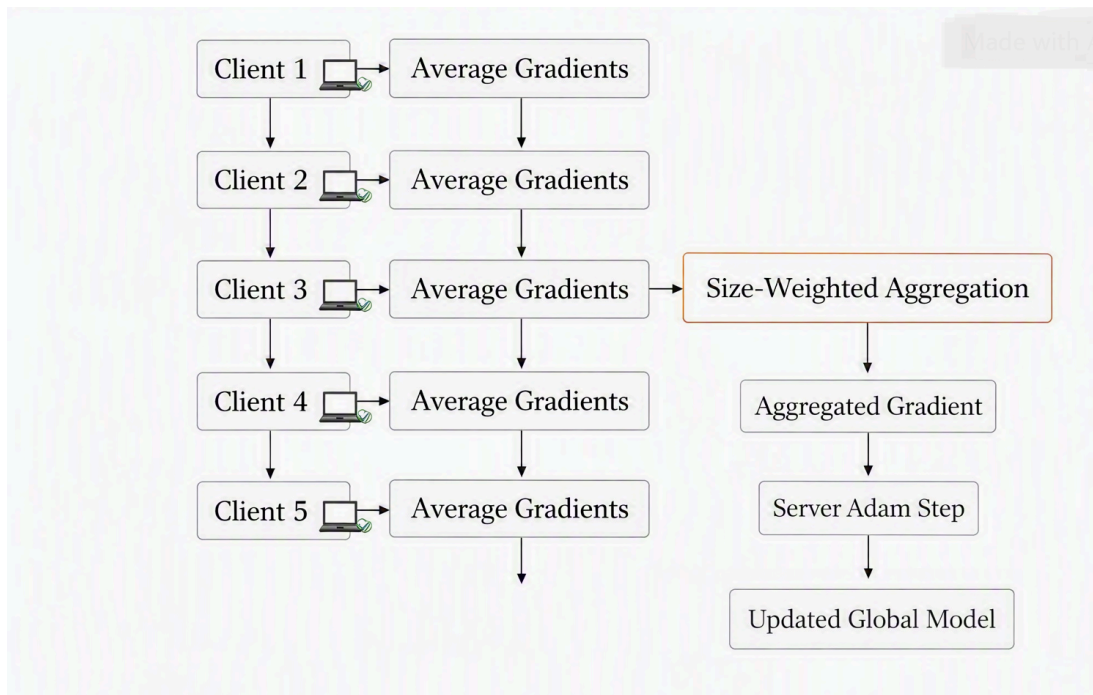


Fig 3.9: Gradient-Based FedAvg Process

Batch Normalization State Aggregation

CNN and Transformer U-Net architectures have Batch Normalization layers. The Batch Normalization running stats are not back-propagated gradients. Therefore, the process

allows the global model to maintain aggregated Batch Normalization statistics across the participating clients.

Federated Communication Rounds

The experiment conducted on segmentation is repeated over three federated communication rounds, where the global model is gradually updated from the distributed clients. Each round is an integrated series of operations which will synchronise local and global learning processes.

The following steps are performed on every round of the communication:

1. Global Model Retrieval: The model parameters are retrieved from the central server.
2. GlobalModelDistribution: All five participating clients receive the global model state.
3. Local Training: Two local epochs of training are done on the specific data set of each client with the specified optimiser and loss functions.
4. Gradient Computation: TensorFlow automatically differentiates to compute local gradients by backpropagation.
5. Batch Normalization State Collection: Client's batch normalization statistic (mean and variance) are collected to ensure consistency in distributed training environment.
6. Gradient Aggregation: The server aggregates gradients received based on a size weighted scheme of its clients' datasets.
7. Global Optimization: The aggregated gradient is passed to the global model with the Adam optimizer, and the model is updated with the new parameters.
8. Collecting the Batch Normalization States: The Batch Normalization states collected are aggregated to synchronize the behavior of Batch Normalization across clients.
9. Validation and Evaluation: The updated model is tested on the validation set to check the segmentation performance.
10. Round Timing: It is the time taken for each communication round to be executed and the time is logged for performance analysis.

Parameter	Setting
Number of clients	5
Communication rounds	3
Local epochs	2
Batch size	16
Learning rate	0.0001
Aggregation	Size-weighted gradient aggregation
Server optimizer	Adam
Client optimizer	Adam
Client partition executed	IID
Dirichlet α	0.5
Model variants	CNN U-Net, Transformer U-Net

Table 3.8: Federated Training Configuration

Evaluation Metrics

Five common metrics are used to test the performance of the segmentation models: Dice coefficient, Intersection over Union (IoU), Precision, Recall, and F1-score. These metrics

overall assess the accuracy at the pixel level and intersection at the region level between the predicted tumor masks and the ground truth masks.

- **Dice Coefficient**

The Dice coefficient measures how similar the segmentation and the ground truth mask are:

$$Dice = \frac{2TP}{2TP + FP + FN} \quad \text{xii}$$

The Dice value is a key measure for the quality of segmentation with a higher value suggesting greater similarity between the predicted and actual tumor regions.

- **Intersection over Union (IoU)**

The IoU is the ratio of the intersection over the union of predicted and ground truth tumor pixels:

$$IoU = \frac{TP}{TP + FP + FN} \quad \text{xiii}$$

This measure complements Dice by focusing on the amount of correctly predicted pixels in the intersection of both sets.

- **Precision**

$$Precision = \frac{TP}{TP + FP}$$

Precision measures the percentage of the correct prediction of tumor pixels, suggesting the accuracy of positive predictions.

- **Recall**

$$Recall = \frac{TP}{TP + FN} \quad \text{xiv}$$

Recall is the percentage of true tumor pixels that are correctly classified, giving insight into the sensitivity of the segmentation model.

- **F1-Score**

The F1-score is the harmonic mean of Precision and Recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad \text{xv}$$

This is a metric that integrates the trade-off between Precision and Recall, as a comprehensive measure of segmentation performance. The segmentation models are assessed using five typical metrics such as Dice coefficient, Intersection over Union (IoU), Precision, Recall, and F1-score. These metrics give a full picture of the accuracy, overlap and reliability of the segments.

Metric	Measurement
Dice	Segmentation overlap between prediction and ground truth
IoU	Intersection relative to union of predicted and actual pixels
Precision	Correctness of positive predictions (tumor pixels)
Recall	Ability to detect actual tumor pixels
F1-score	Balance between Precision and Recall

Table 3.9: Evaluation Metrics

- **Thresholding for Binary Masks**

Creating Binary Masks from thresholding. Creating Binary Masks using Thresholding. Predicted probability maps are then thresholded to produce a binary segmentation mask for the final test evaluation with a fixed threshold:

$$threshold = 0.5 \quad \text{xvi}$$

Those pixels classified as tumour are those for which there is a greater than 50% probability of being tumour and those below 50% are classified as background. This binarization step helps to have a consistent metric computation for all the evaluation experiments.

Communication Cost Analysis

Besides the segmentation performance, the implementation also assesses the communication needs of federated learning. Analysis of communication is vital to understanding the efficiency and scalability of distributed training systems.

The models are implemented in float32 representation and each model parameter is assumed to be 4 bytes. The communication calculation takes the following into consideration:

- Number of model parameters transmitted
- Number of participating clients
- Number of communication rounds
- Client-to-server communication (upload)
- Server-to-client communication (download)

The total communication volume is expressed as:

$$C = P \times 4 \times K \times R \times 2 \quad \text{xvii}$$

where:

- P = number of transmitted parameters,
- 4 = bytes per float32 parameter,
- K = number of clients,
- R = number of communication rounds,
- 2 = factor accounting for both upload and download communication.

This formulation reflects the "two-way communication overhead" that is inherent to FL.

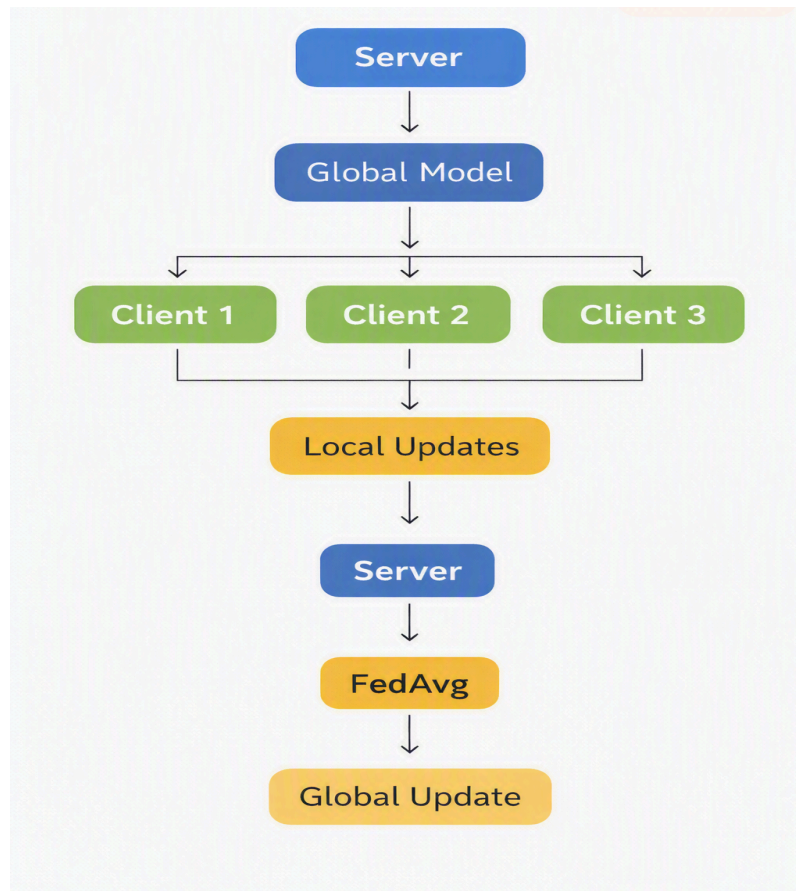


Fig 3.10: Federated Communication Process

3.3 Project Plan

The project consists of a series of stages, each one playing a role in the overall research and implementation process.

❖ Phase 1 – Literature Review and Dataset Collection

A survey of recent research about fake news detection based on deep learning and transformer-based models is conducted. Special attention is paid to hybrid approaches, interpretability techniques (e.g., LIME), and the problems arising from low-resource languages. Meanwhile, the BanFakeNews is obtained from Kaggle and fact-checking portals, which consists of around 50,000 Bangla news articles with two labels authentic and fake.

❖ Phase 2 – Preprocessing and Baseline Model Development

The phase involves preprocessing and baseline modeling. The second phase is to preprocess and develop baseline models. A full text pre-processing pipeline is designed, such as cleaning, Normalization, Tokenization and oversampling and augmentation of class imbalance. Baseline machine learning models are implemented, including Logistic Regression, Naïve Bayes and SVM, while deep

learning baselines are implemented such as BanglaBERT Vanilla, which are used as benchmarks for the development of the hybrid models.

❖ Phase 3 – Ensemble Model Training and Optimization

Focusing on fine tuning transformer based models like Bangla-BERT, Bangla-XLM-RoBERTa and integrating these with machine learning classifiers using ensemble techniques such as soft voting, stacking etc. Methods such as dropout, learning rate scheduling and early stopping are used to ensure stability and generalization.

❖ Phase 4 – Evaluation, Visualization and Explainability

Evaluation of the hybrid model is done against baseline models by accuracy, precision, recall, the F1-score, and AUC. The classification performance is evaluated through visualization techniques such as confusion matrices and ROC curves, and the predictions' interpretability is achieved by identifying important words and contextual focus using LIME and attention heatmaps.

❖ Phase 5 – Report Writing and Documentation

This will be the fifth phase of the activity, focusing on report writing and documentation. Lastly, prepare a detailed thesis report of the overall methodology, experimental results and analysis. Figures, tables, graphs and explainability visualization are created as well as slides, flowcharts and diagrams for the defense presentation. This makes sure that the research is well documented and communicated.

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project. from week 6 to week 23.

Tasks	Weeks																	
	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Phase-1&2																		
Phase- 3																		
Phase- 4																		
Phase- 5																		

Estimated Work Period	
Actual Work Period	

3.5 Summary

This study implements a brain tumor segmentation framework using 2,146 MRI images split into training, validation, and testing sets. Images are resized to 128×128 and annotated into binary masks. Two architectures are compared: a CNN-based U-Net and a Transformer-enhanced U-Net with self-attention. Models are trained with Binary Cross-Entropy + Dice loss using Adam at a learning rate of 1×10^{-4} . Federated learning is simulated with five IID clients, each running local training before gradient aggregation and server-side updates. Performance is evaluated using Dice, IoU, Precision, Recall, and F1-score, alongside communication and execution analysis. While Non-IID client generation is implemented, the reported experiments focus on IID partitioning. Overall, the methodology establishes a controlled setup to compare CNN and Transformer U-Net models under centralized and federated learning.

Chapter 4

Implementation and Results

This chapter presents the implementation methodology of the proposed system, the environment and tools used, the model training–testing process, and the final results. It shows how data processing, model building, and evaluation are performed step by step, and the effectiveness of the proposed method is evaluated through various performance metrics.

4.1 Environment Setup

The proposed brain tumor segmentation framework was implemented using a Python-based deep learning environment. The complete experimental pipeline was developed and executed in a Kaggle-based computational environment, where the dataset and training resources were accessed through the notebook environment.

The implementation consists of several major components, including dataset preparation, image and annotation processing, segmentation model development, centralized training, federated learning, performance evaluation, visualization, and communication-cost analysis.

TensorFlow and Keras were used as the primary deep learning frameworks for constructing and training the CNN-based U-Net and Transformer-based U-Net architectures. NumPy was used for numerical array operations, while Pandas was used for organizing experimental results. OpenCV was used for image processing and rasterization of polygon-based tumor annotations. Scikit-learn was used for calculating evaluation metrics, while Matplotlib and Seaborn were used for generating performance visualizations.

Component	Implementation
Programming Language	Python
Development Environment	Kaggle Notebook
Deep Learning Framework	TensorFlow / Keras
Numerical Computing	NumPy
Data Processing	Pandas
Image Processing	OpenCV
Machine Learning Utilities	Scikit-learn
Visualization	Matplotlib, Seaborn
Annotation Format	COCO JSON
Optimizer	Adam

Table 4.1: Software and Development Environment

The images were split into training (1,502 images), validation (429 images), and testing (215 images) sets. The implementation also conducted the dataset quality checking. No corrupted, duplicate or blank images were excluded in the pre-processing stage.

Implementation of Tumor Mask Generation

The dataset annotations were made with the COCO annotation format. Implementation reads the annotation json files and extracts polygon coordinates that represent the regions of the tumors. For every image, an empty mask is initially created. The tumor polygon coordinates are then converted to pixel regions using OpenCV. A polygon is used to fill the area and create the ground-truth binary mask. A mask is then created, and resized to 128 x 128 pixels with nearest-neighbor interpolation. This maintains the segmented nature of the segmentation labels. The implementation is different for the following two types of images: There are two types of images, and the implementation is different:

Dataset Split	Total Images	Images with Non-Empty Tumor Mask
Training	1,502	730
Validation	429	219
Testing	215	97

Table 4.2: Tumor Mask Distribution

Of the 1,502 training images, 730 have nonempty tumor masks. This means that this dataset includes samples from tumors and samples from background.

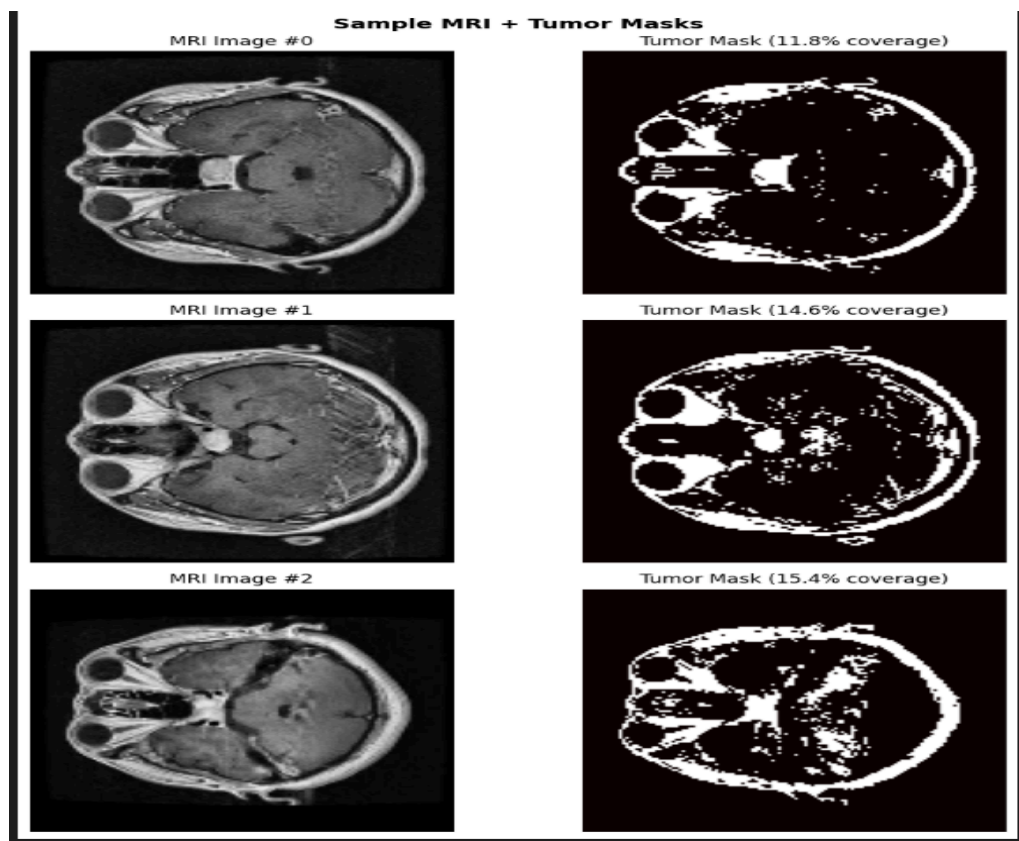


Figure 4.1: Example of MRI Image and Generated Tumor Mask

One representative MRI image and one corresponding binary mask of the tumor should be displayed next to each other.

4.2 Testing and Evaluation/Performance/Comparative Analysis

This system was tested in the centralized and federated learning configurations. Two ways of segmentation was taken into consideration:

1. CNN-based U-Net
2. Transformer-based U-Net

The CNN U-Net is the conventional baseline of encoder-decoder network, while the Transformer U-Net adds a Multi-Head Self-Attention mechanism at the bottleneck.

The models were evaluated using Dice coefficient, Intersection over Union (IoU), Precision, Recall, and F1-score.

These metrics were selected because segmentation performance cannot be adequately represented by classification accuracy alone. Dice and IoU directly measure the overlap between the predicted tumor region and the ground-truth segmentation.

CNN U-Net Implementation

The CNN U-Net was implemented using an encoder-decoder architecture with skip connections.

The encoder progressively extracts hierarchical image features using convolutional layers. The decoder reconstructs the spatial resolution of the feature representation while receiving corresponding encoder features through skip connections.

The number of filters increases progressively from 32 to 512 in the encoder and decreases from 256 to 32 in the decoder.

Component	Configuration
Input	$128 \times 128 \times 3$
Encoder Filters	32, 64, 128, 256
Bottleneck Filters	512
Decoder Filters	256, 128, 64, 32
Convolution Kernel	3×3
Output Layer	1×1 Convolution
Output Activation	Sigmoid
Normalization	Batch Normalization
Pooling	Max Pooling
Upsampling	Transposed Convolution

Table 4.3: CNN U-Net Configuration

The final sigmoid layer generates a probability map for the tumor class. A threshold of 0.5 is subsequently applied to convert the probability map into a binary segmentation mask.

Transformer U-Net Implementation

The Transformer U-Net uses the same general encoder-decoder structure as the CNN U-Net.

The major architectural difference occurs at the bottleneck.

The bottleneck feature representation is transformed into a sequence of spatial tokens and processed using Multi-Head Self-Attention.

The implemented Transformer block uses four attention heads, a key dimension of 32, and a feed-forward dimension of 512.

Parameter	Value
Attention Heads	4
Key Dimension	32
Feed-Forward Dimension	512
Normalization	Layer Normalization
Residual Connections	Yes
Bottleneck Channels	512
Output Activation	Sigmoid

Table 4.4: Transformer U-Net Configuration

The purpose of introducing self-attention is to allow the model to capture relationships between spatial features beyond the local receptive fields typically emphasized by convolutional operations.

Training Configuration

Both segmentation architectures were trained using the Adam optimizer and the combined Binary Cross-Entropy and Dice loss.

Input Resolution	128 × 128
Batch Size	16
Learning Rate	0.0001
Optimizer	Adam
Loss Function	Binary Cross-Entropy + Dice Loss
Maximum Centralized Epochs	15
FL Local Epochs	2
FL Communication Rounds	3
Number of FL Clients	5
Prediction Threshold	0.5

Table 4.6: Model Training Parameters

Early stopping and learning-rate reduction were incorporated into centralized training to reduce unnecessary training when validation performance stopped improving.

Centralized Learning Results

A centralized experiment was conducted prior to federated training and gave a baseline for comparison. The CNN U-Net test Dice coefficient was 0.1811 and an IoU of 0.0996. Its Precision was 0.7418, Recall was 0.8485, and F1-score was 0.7916. The achieved test Dice coefficient and IoU for Transformer U-Net are 0.1695 and 0.0926 respectively. Its Precision was 0.7025, Recall was 0.8577, and F1-score was 0.7723.

Model	Dice	IoU	Precision	Recall	F1
CNN U-Net	0.1811	0.0996	0.7418	0.8485	0.7916
Transformer U-Net	0.1695	0.0926	0.7025	0.8577	0.7723

Table 4.7:Centralized Segmentation Results

The CNN U-Net outperformed the Transformer U-Net in terms of Dice and IoU scores. the centralized experiment. It also performed a bit better in terms of F1 score. In contrast, a higher Recall value was obtained from the Transformer U-Net. This To assess this, the model Transformer was tested and it was found that the model identified a larger The result was a lower overlap between the result and ground truth, but a higher proportion of tumor pixels. segmentation.

Federated Learning Implementation

The federated learning experiment took place in a decentralized health care system.The federated learning experiment is conducted in a decentralized healthcare system. Model Dice IoU Precision Recall F1

- **Federated Transformer & CNN U-Net Results**

The federated CNN U-Net was evaluated on the held-out test dataset after federated training.

The model obtained a Dice coefficient of 0.1570 and an IoU of 0.0852. The Precision, Recall, and F1-score were 0.7593, 0.8327, and 0.7943, respectively.The results indicate that the federated CNN U-Net retained a relatively high Recall while its Dice and IoU scores were lower than those obtained in the centralized CNN experiment.

The federated Transformer U-Net achieved a test Dice coefficient of 0.1429 and an IoU of 0.0770. The model obtained Precision of 0.7664, Recall of 0.7581, and F1-score of 0.7622.

Model	Dice	IoU	Precision	Recall	F1
Federated CNN U-Net	0.1570	0.0852	0.7593	0.8327	0.7943
Federated Transformer U-Net	0.1429	0.0770	0.7664	0.7581	0.7622

Table 3.11: Federated Segmentation Results

The Transformer model achieved the highest precision among the four tested configurations, but its Dice, IoU, Recall, and F1-score were lower than those of the federated CNN U-Net.

4.3 Results and Discussion

The experimental results demonstrate that the performance of the implemented segmentation models depends on both the architecture and the training strategy.

The CNN U-Net achieved the strongest Dice and IoU performance in the centralized experiment. This result is consistent with the general suitability of U-Net-style encoder-decoder architectures for semantic segmentation, where spatial information from the encoder can be transferred to the decoder through skip connections. The literature review similarly identifies CNN and U-Net architectures as important approaches for automated brain tumor segmentation.

The Transformer U-Net was introduced to improve the ability of the model to capture global

relationships among spatial features. However, the current experimental configuration did not produce higher overall segmentation overlap than the CNN U-Net. One possible interpretation is that the relatively small input resolution of 128×128 and the available training configuration may not provide enough information for the Transformer component to demonstrate a clear advantage.

The results also demonstrate a difference between Recall and overlap-based metrics. The centralized Transformer U-Net achieved the highest Recall of 0.8577, while its Dice and IoU were lower than those of the CNN U-Net. This indicates that the model detected a considerable amount of tumor pixels but did not achieve equally strong spatial agreement with the ground-truth masks.

The federated CNN U-Net obtained a Dice score of 0.1570 compared with 0.1811 for centralized CNN U-Net. Similarly, the federated Transformer U-Net obtained a Dice score of 0.1429 compared with 0.1695 for its centralized counterpart.

Thus, with the current experimental setup, the federated training approach led to an improvement in the following results: Lower segmentation overlap as compared to centralized training. This difference may be attributed After finishing distributed training, with only a few rounds of communication, and local training The use of simulated client partitions and the configuration. In federated learning, however, there is a key architectural benefit to In healthcare environments: raw training images may be kept in the respective client. environments, not directly merged into a single, centralized environment. This According to the literature reviewed, there is strong evidence of motivation being supported by the reviewed literature, and the reviewed literature identified federated. The use of learning as a potential tool for collaborative medical imaging with privacy constraints. data-sharing constraints. The literature also shows that the federated brain tumor segmentation can attain Competitive results in more extensive experimental settings. For example, previous studies have investigated U-Net-based FL, privacy-preserving techniques, differential More privacy and larger multi-institutional datasets. Hence, the relatively poor Dice scores achieved as a result of the present implementation should not be taken to imply that federated learning is a bad choice over centralized learning. Instead, they are just limited to the specific dataset, architecture, number of Parameters of the clients, number of communication rounds, local epochs, training configuration, etc. in this experiment. Yet another point to note is that the experiment conducted at present is federated with IID client partitioning. The implementation includes the ability to support Non-IID data The aforementioned experiment was carried out in distribution based approach which is Dirichlet. With the IID configuration. The conclusions about Non-IID federated datasets were drawn from the above results. You can not take the current results as evidence for performance. The client-wise results indicate that the CNN U-Net had higher Dice scores in all cases. all five clients than the Transformer U-Net. This provides additional evidence the The current experimental set up was better suited to CNN architecture.

In general the experimental results are as follows:

1. CNN U-Net outperformed the rest with its highest Dice and IoU.
2. CNN U-Net achieved the best overall segmentation overlap compared to Transformer U-Net.
3. Transformer U-Net outperformed the other models in terms of centralized Recall.
4. For Dice, IoU, Federated CNN U-Net performed better than Federate Transformer

U-Net, as shown in Recall, and F1-score.

5. Federated Transformer U-Net obtained the best precision in the tested. models.
6. Both federated models had worse Dice at the end of the test than their centralized counterparts.
7. The performance of CNN U-Net was also better than all other five simulated when it came to Dice values in terms of client-wise performance. clients.
8. In the current federated experiment, IID client partitioning was used.
9. The reported did not include the following implemented Non-IID mechanism: experimental results.
10. Collaborative training is shown to be feasible in the federated framework. Without having to merge the training images for distribution.

4.4 Summary

This chapter presented the implementation and comparative evaluation of the proposed brain tumor segmentation framework using 2,146 MRI images. Two architectures, CNN U-Net and Transformer U-Net, were tested under centralized and federated learning. The centralized CNN U-Net achieved the best Dice (0.1811) and IoU (0.0996), while the Transformer U-Net showed stronger Recall (0.8577). In federated experiments with five IID clients, both models showed reduced overlap compared to centralized training, but demonstrated the feasibility of distributed learning. Overall, CNN U-Net provided better segmentation performance, while Transformer U-Net showed strengths in Recall and Precision. These findings establish the basis for conclusions, limitations, and future improvements discussed in the next chapter.

Chapter 5

Engineering Standards and Design Challenges

This chapter introduces the engineering standards, design challenges, social and ethical issues, project management challenges, and implementation strategies for the proposed cancer diagnostic system. It has been developed while complying with software, hardware, data security, communication and clinical guidelines.

5.1 Compliance with the Standards

The proposed framework for brain tumor segmentation was designed as a medical image analysis system that takes the concept of deep learning and federated learning. The focus on compliance is primarily on quality of software, reliability of the computation, safe handling of medical data, reproducibility and communication between the simulated federated clients and the global server, due to the experimental and research nature of the work.

Implementation is done using the Python environment with machine learning libraries TensorFlow/Keras, NumPy, Pandas, OpenCV, Scikit-learn, Matplotlib, and related libraries. It comprises centralized model training, federated client simulation, CNN-based U-Net, Transformer-based U-Net, COCO annotation processing, binary tumor-mask generation, and performance evaluation.

The system is not currently used as a clinical medical device, yet the design has several good engineering practices with regard to modular software development, reproducibility, privacy-aware data processing, and reliable experimental evaluation.

5.1.1 Software Standards

The software development adheres to structured programming principles and machine learning development principles. The entire pipeline is broken down into individual functions: Dataset Loading, Annotation Processing, Preprocessing, Model Construction, Centralized Training, Federated Client, Local Training, Aggregation, Evaluation, and Visualization. Python is the main programming language due to its extensive support for scientific computing, machine learning and medical image processing. Model implementation and optimization is carried out using TensorFlow and Keras; image annotations are processed by OpenCV and segmentation masks are generated.

The implementation follows several software quality principles as follows:

- Modular design of preprocessing, model building, training, and evaluation functions.
- Separation between centralized and federated experiments.
- Consistent preprocessing for training, validation, and testing data.
- Use of fixed image dimensions to maintain model compatibility.
- Use of standard evaluation metrics for reproducibility.

- Consistent optimizer and loss configurations for model comparison.
- Recording of model performance across different experimental settings.
- Use of structured data formats such as COCO JSON for annotation processing.

The experimental implementation also takes care of reproducibility by means of uniform partitions of the dataset and training setup. These help to minimise the variation in implementation and enhance the consistency of comparative experiments.

Engineering Practice	Application in the Project
Modular programming	Separate functions for preprocessing, training, FL, and evaluation
Data consistency	Fixed 128×128 image resolution
Standardized annotations	COCO JSON processing
Model reproducibility	Defined architecture and training parameters
Performance validation	Dice, IoU, Precision, Recall, F1-score
Model comparison	Centralized vs. Federated and CNN vs. Transformer
Code portability	Python/TensorFlow-based implementation
Experiment documentation	Results stored and visualized

Table 5.1: Software Engineering Practices Used in the Project

The software engineering practices utilized in the project are listed in Table 5.1. The system is not yet formally certified to a clinical software standard, it is a research prototype and not a medical device.

5.1.2 Hardware Standards

It is mainly software based and not reliant on dedicated medical hardware. The experiments are run on a cloud-based Kaggle notebook environment. The project's computational needs are to be:

- Preprocessing and data manipulation using CPU.
- GPU acceleration for end-to-end deep neural network training (if available).
- Memory to store image arrays and intermediate tensors for the system.
- Storage of MRI data, annotation files, model weights and results.

To reduce computational cost, but still contain enough information for the experimental segmentation task, the models were trained with 128×128 input images.

The federated learning environment is emulated on the same computation platform. So, there were not five separate machines or hospitals used in the experiment. Instead, it is assumed that each Client is a separate data partition and a separate training process.

Component	Requirement
Processor	CPU-compatible environment
Accelerator	GPU recommended for deep learning
Memory	Sufficient RAM for image arrays and models
Storage	Dataset, model weights, and outputs
Client Environment	Five simulated clients
Server Environment	Simulated global aggregation server
Input Resolution	128 × 128 pixels

Table 5.2: Hardware and Computational Requirements

For future real-world implementation, separate secure computing environments at participating healthcare institutions would be required.

5.1.3 Communication Standards

Communication is an important component of federated learning because the global model and client updates must be exchanged repeatedly during training.

In the implemented system, communication is simulated within the same notebook environment rather than through a real network protocol. During each federated round, the global model parameters are distributed conceptually to the participating clients. Each client performs local training and returns its locally averaged gradient information. The server then performs size-weighted aggregation.

The communication flow is:

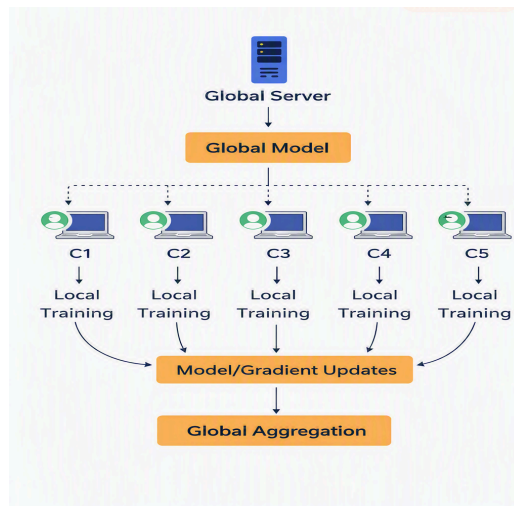


Fig 5.1 : Communication of client

There is no real-world networking protocol like HTTPS, TLS, MQTT, gRPC or REST based federated communication defined or deployed in the current implementation. Thus, these protocols should not be endorsed as being implemented. If it were to be deployed in the future, an encrypted communication would need to be used to safeguard model data exchanged between healthcare institutions.

5.2 Impact on Society, Environment and Sustainability

The proposed research may have implications to go beyond the scope of model performance. Federated learning in medical image analysis could potentially impact access to healthcare, privacy, institutional partnerships, computational resource consumption, and the responsible use of artificial intelligence.

5.2.1 Impact on Life

Brain tumor segmentation can be automatically performed, which may help doctors identify and visualize the regions of the brain tumor in MRI images.

Manual segmentation of tumors is time-consuming, especially when a large amount of medical images need to be reviewed. An automatic segmentation system could help alleviate repetitive tasks in image analysis and offer the clinician another tool for computation. Potential benefits include:

- Quicker initial processing of MRI images.
- Possibility of automated identification of tumor regions.
- Support for treatment planning.
- Support for tracking tumour growth.
- Minimization of repetitive manual effort to segment.
- Better availability of computer assisted medical image analysis.

The current system is an experimental research system, and cannot replace professional medical diagnosis. The model's prediction should not be used to make a final clinical decision, but as a computational aid.

5.2.2 Impact on Society & Environment

Collaboration between several health care institutions can potentially be facilitated using federated learning, where they do not have to centralize all imaging data from patients. It could be helpful for society in removing obstacles to joint research activities in the field of artificial intelligence. Federated learning can offer a way to share local datasets with a shared model even when smaller medical institutions have limited data sets.

There are potential societal benefits that include:

- Increased collaboration between medical institutions.
- Decreased reliance on central medical data stores.
- Enhancing use of distributed datasets.
- Improved access for small data institutions to research using AI.
- Improvement of more varied medical AI systems.

Deep learning training consumes computational resources and electrical power, with an environmental perspective. Moreover, multiple clients repeatedly update local models, which can further boost computation in federated learning. The present implementation decreases part of the computational requirement by resizing images to 128×128 and the number of communication rounds to a limited number.

On a wide scale, however, the scaling up of the model size, the frequency of local training and the number of communication rounds, as well as the size of the servers, would have to be optimized to avoid unnecessary computations and energy usage.

Area	Positive Impact	Potential Challenge
Healthcare	Automated segmentation support	Prediction errors
Privacy	Raw data remain distributed	Model updates may still expose information
Collaboration	Multi-client learning	Coordination complexity
Computation	Shared learning	Higher distributed computation
Environment	Efficient image resizing	Energy consumption from repeated training
Accessibility	AI support across institutions	Infrastructure requirements

Table 5.3: Positive and Negative Impacts

5.2.3 Ethical Aspects

Ethical concern: don't put the raw MRI data in the central place. Local data is stored on the client and only model updates are aggregated by the server. Federated Learning is not fully private (limitation). No formal privacy mechanisms are in place:

- Differential Privacy
- Homomorphic Encryption
- Secure Aggregation
- Trusted Execution Environments

Challenges: bias in models, incorrect predictions, representativeness of datasets, accountability. Principles for deployment:

- Patient information is subject to permission to use.
- No personally identifiable information will be included.
- However, AI should not be used as a replacement to clinical judgment.
- Before use in the clinic, the quality of the water will be validated independently.
- Limits of the model are clearly communicated
- To include a variety of populations to avoid bias.
- Secure access to medical AI systems

5.2.4 Sustainability Plan

- To use lightweight segmentation architectures, if applicable
- Minimize the number of unnecessary federated communications rounds
- Run local epochs and client participation optimally.
- Transfer learning using trained model weights.
- Ensure consistent preprocessing pipeline.
- Keep your model checkpoints to not re-train.
- Only use scalable cloud GPU resources or institutional GPU resources when

necessary.

- Support new clients without redesigning the complete system
- Retrain models with new medical data from time to time
- Track model performance to see if model degradation occurs over time

5.3 Project Management and Financial Analysis

An understanding of project management and financial analysis. The project has been well designed and managed in accordance with the cost conscious planning. The following is a comprehensive financial analysis:

Table 5.4: Project Management and Financial Analysis Table

Components	Estimated Cost (BDT)
Internet and Cloud Subscription (Google Colab Pro+ / Basic HPC Use)	2,500 – 3,000
Data Collection and Preparation	1000 - 2,000
Contingency (10% Buffer)	500
Miscellaneous (Printing, Backup Storage)	1,500–2,000
Total Estimated Cost	5500 – 7500 BDT

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

Table 5.5: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stakeholder Involvement	EP7 Interdependence
✓	✓	✓	✓			✓

EP1 – Depth of Knowledge

Required ML, DL, medical imaging, federated learning. Implemented CNN U-Net and Transformer U-Net.

EP2 – Conflicting Requirements

Trade-off: performance vs. computation, complexity vs. cost. Balance needed across privacy, training time, communication

EP3 – Depth of Analysis

Dataset preprocessing and mask generation. Metrics: Dice, IoU, Precision, Recall, F1. Compared centralized vs. federated models.

EP4 – Familiarity of Issues

Standard methods: U-Net, CNNs, Adam, BCE, Dice loss. Advanced: federated training, aggregation, Transformer attention

EP5 – Extent of Applicable Codes

Guided by HIPAA/GDPR principles. Prototype not certified, but privacy-aware design used

EP6 – Stakeholder Involvement

Patients, doctors, hospitals, researchers, engineers, regulators. Priorities: reliability, privacy, responsible data use

EP7 – Interdependence

Dataset → preprocessing → training → performance. Federated learning links local training, aggregation, communication, evaluation

Mapping with Knowledge Profile

Table 5.6: Mapping with knowledge Profile.

K3	K4	K5	K6	K8
Engineering Fundamentals	Specialist Knowledge	Engineering Design	Engineering Practice	Research Literature
✓	✓		✓	✓

K3 – Engineering Fundamentals

Basic knowledge of optimization, mathematical modeling, and data communication was used.

K4 – Specialist Knowledge

Deep learning, CNN/ensemble methods, and medical imaging—this specialized knowledge was required.

K5 – Engineering Design

Data preprocessing, model architecture, and evaluation planning—all were done according to design principles.

K6 – Engineering Practice

Python implementation, cloud computing, debugging, experimentation, testing, and result analysis.

K8 – Research Literature

Recent research papers were analyzed to guide the project.

5.4.2 Engineering Activities

Mapping with Complex Engineering Activities

Table 5.4: Mapping with Complex Engineering Activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓		✓	✓

EA1 – Range of Resources

Datasets, models, cloud computing—a wide range of resources were used.

EA2 – Level of Interaction

Interaction exists between dataset preprocessing, model training, federated clients, global aggregation, and evaluation.

EA3 – Innovation

Various data features, metrics, class imbalances, and model tuning—all had to be analyzed.

EA4 – Consequences for society and environment

Social and lifelong impact—where the results are directly applicable to patient care.

EA5 – Familiarity

The project combines established techniques with their integration into a distributed medical segmentation framework.

5.5 Summary

The project addressed engineering, financial, ethical, social, and technical considerations in brain tumor segmentation. It was developed at cost-consciously using open-source tools and cloud resources, with an estimated cost of 5,500–7,500 BDT. The framework integrates deep learning, medical imaging, semantic segmentation, Transformer attention, and federated learning. It demonstrates complex engineering knowledge, stakeholder involvement, privacy concerns, and interdependence of components. Overall, the project applies advanced engineering to healthcare while considering performance, cost, ethics, sustainability, and societal impact.

Chapter 6

Conclusion

6.1 Summary

The present research proposed a deep learning-based brain tumor segmentation framework that automatically detects brain tumors and segments them. With CNN U-Net, Transformer U-Net and Federated Learning. The study used 2,146 MRI images with tumor annotations. Preprocessing the images and The image has been cropped to a 128×128 size and the annotations were translated into a binary image. Masks. Centralized and federated training were explored. In the federated The data set was split across five simulated clients, within the setting. The updates were compiled to form a global model. Experimental results indicated that the CNN U-Net outperformed the other three methods in overall Dice and of the methods in the table above in the model accuracy assessment. Better performance on the IoU than the Transformer U-Net with the tested setting. The centralized models also performed better than their corresponding federated models.

However, the federated experiments demonstrated the feasibility of collaborative model Directly combining client training data is not involved in training. In summary, the research will give a starting point for privacy-preserving segmentation of brain tumors.using federated deep learning.

6.2 Limitation

There are a number of limitations to the study:

- The data set was quite small and from a single source.
- Fine tumor detail could be reduced due to resizing the images to 128×128 pixels. details.
- Only five simulated federated clients were used.
- The federated environment was simulated instead of being deployed throughout real hospitals.
- Limited number of communication rounds in the experiments.
- The real world NON-IID data distribution was not tested.
- Multivariate RDP are employed to safeguard individuals' privacy. Differential Privacy and Secure Multivariate RDP are used to protect individuals' privacy. Aggregation were not implemented.
- The system was not clinically validated by medical people.
- Presentations of advanced Explainable AI techniques were not seamlessly incorporated into the final product.

The limits might influence the generalizability and applicability of the results to real-world situations. reported results.

6.3 Future Work

The proposed framework can be extended in a number of ways for future work. A larger and more diverse multi-institutional dataset can be used to improve generalization. Multimodal Other MRI data, including T1, T2, FLAIR, and contrast-enhanced data can also be added. The federated framework can be extended with more clients, additional communication Realistic Non-IID data distributions and rounds. Advanced privacy techniques such as Differential Privacy and Secure Aggregation may be combined to offer superior data privacy and aggregation. protection.

The advanced Transformer architectures, higher-resolution images, and Explainable AI techniques like Grad-CAM can be explored in future studies. Finally, clinical validation by radiologists and deployment at actual health care institutions would be Significant steps towards practical application.

References

- [1]Ahamed, M. F., Hossain, M. M., Nahiduzzaman, M., Islam, M. R., Islam, M. R., Ahsan, M., & Haider, J. (2023). A review on brain tumor segmentation based on deep learning methods with federated learning techniques. *Computerized Medical Imaging and Graphics*, 110, 102313.
- [2] Ullah, F., Nadeem, M., Abrar, M., Amin, F., Salam, A., & Khan, S. (2023). Enhancing brain tumor segmentation accuracy through scalable federated learning with advanced data privacy and security measures. *Mathematics*, 11(19), 4189.
- [3] Yahiaoui, M. E., Derdour, M., Abdulghafor, R., Turaev, S., Gasmi, M., Bennour, A., ... & Sarem, M. A. (2024). Federated learning with privacy preserving for multi-institutional three-dimensional brain tumor segmentation. *Diagnostics*, 14(24), 2891.
- [4] Biratu, E. S., Schwenker, F., Ayano, Y. M., & Debelee, T. G. (2021). A survey of brain tumor segmentation and classification algorithms. *Journal of Imaging*, 7(9), 179.
- [5] Wen, J., Li, X., Ye, X., Li, X., & Mao, H. (2025). A highly generalized federated learning algorithm for brain tumor segmentation. *Scientific Reports*, 15(1), 21053.
- [6] Naeem, A., Anees, T., Naqvi, R. A., & Loh, W. K. (2022). A comprehensive analysis of recent deep and federated-learning-based methodologies for brain tumor diagnosis. *Journal of Personalized Medicine*, 12(2), 275.
- [7] Lee, E. H., Han, M., Wright, J., Kuwabara, M., Mevorach, J., Fu, G., ... & Yeom, K. W. (2024). An international study presenting a federated learning AI platform for pediatric brain tumors. *Nature communications*, 15(1), 7615.
- [8] Zhou, L., Wang, M., & Zhou, N. (2024). Distributed federated learning-based deep learning model for privacy mri brain tumor detection. *arXiv preprint arXiv:2404.10026*.
- [9] Onaizah, A. N., Xia, Y., & Hussain, K. (2025). FL-SiCNN: An improved brain tumor diagnosis using siamese convolutional neural network in a peer-to-peer federated learning approach. *Alexandria Engineering Journal*, 114, 1-11.
- [10] Albalawi, E., TR, M., Thakur, A., Kumar, V. V., Gupta, M., Khan, S. B., & Almusharraf, A. (2024). Integrated approach of federated learning with transfer learning for classification and diagnosis of brain tumor. *BMC medical imaging*, 24(1), 110.
- [11] Nazir, S., & Kaleem, M. (2023). Federated learning for medical image analysis with deep neural networks. *Diagnostics*, 13(9), 1532.
- [12] Sandhu, S. S., Gorji, H. T., Tavakolian, P., Tavakolian, K., & Akhbardeh, A. (2023). Medical imaging applications of federated learning. *Diagnostics*, 13(19), 3140.

- [13] Bai, L., Wang, D., Wang, H., Barnett, M., Cabezas, M., Cai, W., ... & Wang, C. (2024). Improving multiple sclerosis lesion segmentation across clinical sites: A federated learning approach with noise-resilient training. *Artificial Intelligence in Medicine*, 152, 102872.
- [14] Shiri, I., Sadr, A. V., Amini, M., Salimi, Y., Sanaat, A., Akhavanallaf, A., ... & Zaidi, H. (2022). Decentralized distributed multi-institutional PET image segmentation using a federated deep learning framework. *Clinical Nuclear Medicine*, 47(7), 606-617.
- [15] Kazerooni, A. F., Khalili, N., Liu, X., Haldar, D., Jiang, Z., Anwar, S. M., ... & Linguraru, M. G. (2024). The brain tumor segmentation (brats) challenge 2023: Focus on pediatrics (cbtbn-connect-dipgr-asnr-miccai brats-peds). *ArXiv*, arXiv-2305.
- [16] Bakas, S., Reyes, M., Jakab, A., Bauer, S., Rempfler, M., Crimi, A., ... & Jambawalikar, S. R. (2018). Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the BRATS challenge. *arXiv preprint arXiv:1811.02629*.
- [17] Ranjbarzadeh, R., Bagherian Kasgari, A., Jafarzadeh Ghouschi, S., Anari, S., Naseri, M., & Bendeche, M. (2021). Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities brain images. *Scientific reports*, 11(1), 10930.
- [18] Jiang, Y., Zhang, Y., Lin, X., Dong, J., Cheng, T., & Liang, J. (2022). SwinBTS: A method for 3D multimodal brain tumor segmentation using swin transformer. *Brain sciences*, 12(6), 797.
- [19] Farhan, A. S., Khalid, M., & Manzoor, U. (2025). XAI-MRI: an ensemble dual-modality approach for 3D brain tumor segmentation using magnetic resonance imaging. *Frontiers in Artificial Intelligence*, 8, 1525240.
- [20] Alphonse, S., Mathew, F., Dhanush, K., & Dinesh, V. (2025). Federated learning with integrated attention multiscale model for brain tumor segmentation. *Scientific Reports*, 15(1), 11889.